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A game-theoretic framework for optimizing supply chain coordination and production

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ABSTRACT

This research introduces a groundbreaking competition concept for supply chains, utilizing the Stackelberg game method to address internal entity interactions. In practical scenarios, chain components often partially cooperate, prioritizing individual benefits without a holistic understanding of the entire chain and market dynamics. Achieving complete chain coordination is challenging, expensive, and requires high-level agreement. Our study presents a simultaneous competition model for two supply chains and their internal entities, considering heterogeneous customers in price and time-sensitive classes. Each chain serves regular and special customers with varied delivery times and pricing. This research aims to investigate how competition among supply chains under various conditions impacts metrics like performance, market share and profits. These conditions include collaboration strategy (Centralized or Decentralized Structure) and production approach (Shared or Dedicated Capacity for specific customers). We employed scenario analysis with the Stackelberg Game framework to study strategic and policy choices' impact on supply chain conditions. We identified 10 distinct scenarios for analysis. Using the Stackelberg model, we iteratively solved the developed models until they reached equilibrium in price and delivery time. Our findings suggest that chains benefit more from a cooperative strategy with a Centralized Structure. Market behavior influences the chosen production approach, where adopting a dedicated capacity policy can lead to increased market share and profits if the market leader does so. Alternative strategies result in competitive stances and reduced returns for both chains.

1. Introduction

In today's competitive business landscape, supply chains seek to enhance market share and profitability by addressing key dimensions such as pricing and delivery time, which significantly influence customer satisfaction [16]. Efficient supply chains prioritize cost reduction, while responsive ones focus on rapid delivery [24]. Some firms, like Amazon, exploit customer heterogeneity by offering premium pricing for faster delivery, thus maximizing consumer surplus [11,28]. However, a lack of coordination between distributors and producers often leads to unmet delivery promises, customer dissatisfaction, and reduced profitability. Greater coordination achieved through either collaboration or competition yields higher overall profits (D. [41]. This complexity is further amplified by diverse customer needs, which can be categorized into high-class (time-sensitive) and low-class (price-sensitive) segments.

This study investigates the competition between two supply chains, each comprising a producer and distributor. Chains differentiate themselves through price and delivery time, with high-class customers prioritizing immediacy and low-class customers favoring affordability. Supply chains adopt either a Centralized Structure, emphasizing collaboration, or a Decentralized Structure, focusing on internal

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competition. To align decision-making, distributors set prices based on market demand, while producers determine delivery times. Research has extensively analyzed centralized strategies [12] and decentralized operations [21,33], highlighting their respective impacts on trust, coordination, and costs. This paper extends these efforts by exploring centralized and decentralized strategies across multiple scenarios using the Stackelberg game model. Assigning one chain as the leader and the other as the follower, the study examines the challenges and opportunities inherent in optimizing production capacity (shared vs. dedicated) to serve distinct customer segments profitably.

Fig. 1 illustrates the competition between two supply chains serving two distinct customer classes. The first chain employs a Centralized Structure, integrating the producer and distributor into a unified entity. It adopts a Dedicated Production Capacity policy, allocating separate capacities for different product types to meet the needs of each customer class.

In contrast, the second chain operates under a Decentralized Structure, where the producer and distributor function independently to maximize their respective profits. This chain uses a Shared Production Capacity approach, leveraging its entire production capacity to produce all product types collectively. Products are then sorted and distributed to customer classes according to their specific requirements.

To summarize the study's key elements and strategies addressing competition and cooperation in supply chain management, a flowchart has been provided in Fig. 2.

Fig. 2 illustrates the structured decision-making process adopted in this study to model strategic interactions between two supply chains. The flowchart begins with identifying the competitive environment and classifying customers into time-sensitive and price-sensitive segments. Based on this classification, each chain selects its structure centralized or decentralized and adopts either a cooperative or competitive strategy. In decentralized settings, the choice between dedicated or shared production capacity significantly impacts performance, leading to the formulation of appropriate competition models. These models are then solved using advanced optimization methods, and outcomes are evaluated through scenario simulation. The framework ultimately guides the extraction of strategic insights and the implementation of optimal supply chain strategies to achieve market equilibrium and profit maximization.

Unlike conventional Stackelberg models that typically address single-layer pricing decisions, the present framework extends the concept to a dual-layered structure. It simultaneously captures intra-chain coordination between manufacturers and distributors and inter-chain competition between rival supply chains. This nested strategic interaction is further enriched by incorporating two distinct customer classes—those sensitive to price and those sensitive to delivery time allowing the model to reflect real-world market heterogeneity. By capturing this combination of internal hierarchy and external rivalry, the study fills a gap in the literature where such multidimensional competition has rarely been modeled in an integrated fashion.

Furthermore, the study evaluates these dynamics under ten well-defined strategic scenarios, spanning centralized, decentralized, and hybrid decision-making architectures. These configurations simulate realistic operational strategies and coordination structures in competitive markets. By embedding this spectrum of possibilities into the Stackelberg framework, the model moves beyond conventional applications and provides a richer, more applicable tool for both academic analysis and managerial decision-making. This layered approach enhances the model's practical relevance and supports strategic evaluation under a variety of competitive conditions.

A standout feature of this research is its focus on pricing and delivery time as distinct decision variables, offering deeper insights into supply chain dynamics. Key contributions include modeling uncertain demand conditions to reflect realistic, stochastic environments, evaluating strategies at both chain and firm levels, and designing optimization algorithms that rapidly identify competitive equilibria. Together, these innovations advance the understanding of supply chain competition and strategic optimization.

The structure of this paper is as follows: Section 2 reviews related literature, emphasizing game theory in supply chain management. Section 3 outlines the study's fundamental assumptions. Section 4 details the modeling framework, including formulation approaches and solution methods. Section 5 discusses and analyzes the results, while Section 6 synthesizes key findings and proposes future research directions.

2. Literature review

The field of supply chain management (SCM) has evolved significantly, driven by dynamic market conditions, technological advancements, and shifting consumer preferences [6,30,15]. To remain competitive in complex global markets, organizations increasingly prioritize strategic coordination of supply chain activities [5]. Within this context, game theory has emerged as a vital tool for analyzing and optimizing supply chain decision-making processes [19] and (Mahmoodi et al., 2024b). It provides a rigorous framework for modeling strategic interactions, enabling a deeper understanding of competitive dynamics, collaboration strategies, and operational efficiencies [40]. Several studies have examined supply chain coordination and competition, emphasizing pricing strategies, delivery time optimization, and decentralized decision-making. For instance, [38] explored how visibility and internal integration impact supply chain responsiveness, while [36] analyzed the effects of customer sensitivity to carbon emissions and delivery time on operational strategies. Additionally, [10] introduced a dynamic competition model that accounts for supply chain stability under evolving market conditions. Prior studies have explored data-driven risk management [34,37], demand forecasting through predictive analytics (Ogunsotoet al., 2025) and optimizing supply chain resilience using dynamic competition models [1]. Our study extends these works by integrating Stackelberg competition theory to analyze strategic interactions, enabling a more robust understanding of pricing and delivery time optimization. This research aligns with the journal's

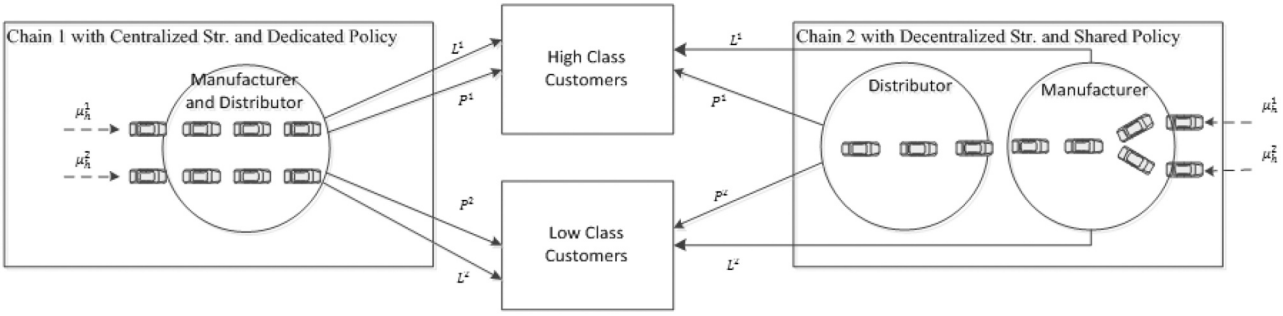


Fig. 1. Chain Competition Example between two chains in the presence of two classes of customers.

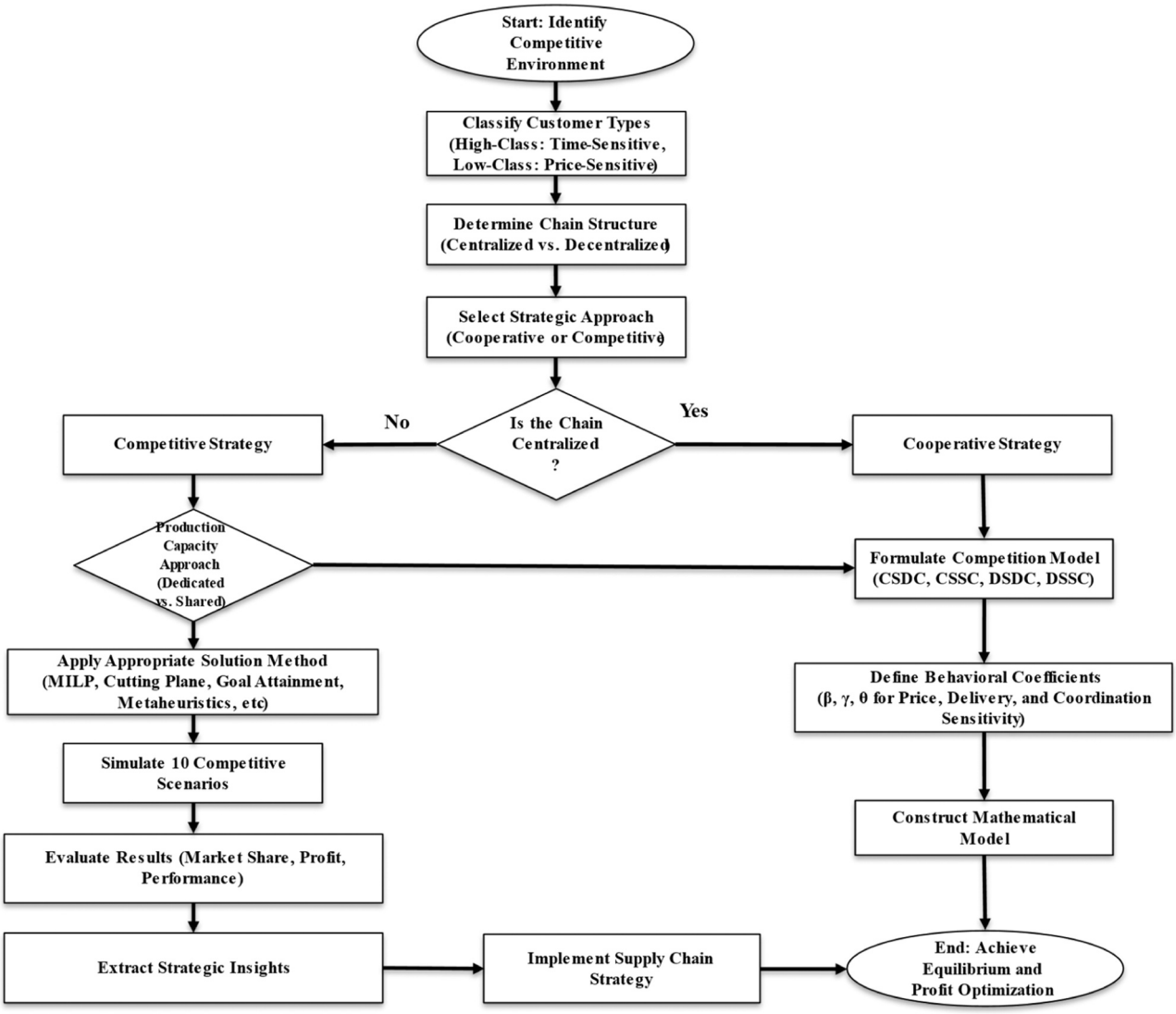


Fig. 2. Flowchart of Strategic Decision-Making in Supply Chain Competition and Coordination.

emphasis on descriptive, predictive, and prescriptive analytics by offering novel game-theoretic approaches to improve operational efficiency and market adaptability. Our research builds upon these contributions by incorporating Stackelberg game theory to analyze competition and cooperation strategies, offering new insights into optimizing supply chain performance.

This literature review explores the application of game theory in SCM, focusing on its role in addressing internal entity interactions and optimizing production decisions. The study introduces a novel competition framework for supply chains, categorized into five distinct competition types. These categories provide a structured foundation for investigating strategic coordination and operational production decisions in dual supply chains, employing the Stackelberg game method as an analytical tool. Cost Competition: Research highlights cost competition as a critical factor, with the lowest marginal cost supply chain often prevailing at economic equilibrium. For instance, Zhang uses spatial price equilibrium theory to show how cooperation among supply chain entities can reduce final costs. Competition models incorporate not only monetary expenses but also inefficiency costs related to delivery and quality. Channel Competition: Several studies examine competition between traditional retail channels and e-tail channels. For example [2,4,29,48] analyze how e-tailers leverage faster delivery and higher prices, while traditional retailers focus on lower prices and slower delivery. These models employ non-cooperative game theory to derive optimal strategies for pricing and lead time. Inventory Competition: Competition in inventory levels has also been extensively studied. Researchers such as [9,25,47] explore how rival retailers compete on price and inventory within supply chains comprising a single supplier and multiple retailers. Using game theory, these studies identify Nash equilibrium points and emphasize the supplier's role in system coordination. Through this review, the application of game theory is shown to provide valuable insights into competition strategies across cost, channel, and inventory dimensions, establishing a strong foundation for further exploration of strategic decision-making in SCM.

2.1. Competition level

2.1.1. Intra-chain competition

This category examines competition among supply chains targeting similar customer segments, focusing on strategic interactions, pricing decisions, and service differentiation strategies to maximize market share and profitability [16,24]. Sustainability has emerged as a critical competitive factor. For instance, Islam, Hassanzadeh Amin, and Wardley proposed a three-step supplier selection and order planning method integrating deep learning for enhanced forecasting accuracy, validated using Canadian meat industry data. [7] utilized Bayesian learning to

optimize sourcing from both reliable expensive suppliers and unreliable cheaper ones, employing dynamic programming and state space reduction techniques for capacity challenges. Similarly [42] developed a sustainable supplier selection model using random forest algorithms and cross-validation, validated in a Chinese forklift company, demonstrating significant accuracy improvements [17] explored duopoly market dynamics, where firms differentiate products based on price and lead time. Their findings suggest price competition reduces price discrimination regardless of market behavior, while competition in lead time impacts product differentiation based on market conditions. Firms with larger markets or cost advantages are advised to maintain greater segmentation between customer groups, offering valuable insights into pricing strategies and capacity management in competitive markets, particularly in make-to-order industries.

2.1.2. Inter-chain competition

This category investigates competition within individual supply chains, focusing on producer-distributor dynamics, resource allocation, market dominance, and customer satisfaction, and their effects on overall chain performance [23,41]. A recent study by [44] introduced a delayed feedback control method to regulate delivery times in intertwined supply chains across countries. The study developed an output dynamic game model, analyzing stability, bifurcations, and chaos through numerical simulations. Findings reveal that higher output adjustment speeds can induce chaos, while tariffs stabilize exporting supply chains. Delayed feedback control emerges as a vital tool for managing chaotic behaviors, maintaining market stability, and optimizing competition strategies [44,43].

2.2. Strategic choices: collaboration vs. competition

This category examines the strategic decisions supply chain components make regarding collaboration or competition, emphasizing the trade-offs between centralized and decentralized decision-making structures. These choices significantly impact coordination, trust, and profitability within the supply chain [26,21,31]. Chen explored decentralized strategies to identify optimal approaches for rival firms, focusing on customer demand sensitivity to price and service levels. The study introduced models of "coopetition" (cooperative competition) for firms with varying service-investment efficiencies, demonstrating that the decision to cooperate or compete—and the extent of cooperation—depends on market competition intensity. This framework provides strategic insights for firms to navigate competition, which can either mitigate or intensify competition across market segments [45].

Conversely [18] used a game-theoretical model to analyze how incumbent and entrant firms in ICT markets decide between technological cooperation (e.g., licensing) and competition (e.g., in-house R&D). Their findings indicate that centralized technological cooperation is preferred under specific conditions, such as favorable technology transfer rates and entrant R&D capabilities. Factors like royalty rates, price competition, and the R&D phase duration also influence these strategic decisions, showing that centralized strategies can maximize social welfare in certain scenarios. Similarly [22] investigated centralized strategies in ICT markets, focusing on licensing practices and technological competition. They found that the equilibrium strategy whether cooperation or competition depends on the interplay of technology transfer rates, R&D phase lengths, and market conditions. Under specific circumstances, centralized approaches enhance social welfare, benefiting both firms and stakeholders.

2.3. Competition and cooperation strategies for customer class differentiation

This research area explores strategies tailored to different customer classes, focusing on how supply chains differentiate offerings based on pricing, delivery time, and service levels to enhance market penetration and customer satisfaction [11] investigated a homogeneous supply chain influenced by consumer sensitivity to carbon emissions and delivery times using Stackelberg game models. The study highlighted how shifts in sensitivity affect operational strategies and introduced a novel two-way cost-sharing contract to counteract free-riding, achieving Pareto improvement across the supply chain. In a dual-channel supply chain [35] examined the impact of establishing virtual showrooms on homogeneous consumer groups: H-consumers with experience costs and L-consumers without virtual showrooming behavior (VSB). Using a Stackelberg game model, the research found that strategies for accommodating or promoting VSB depend on consumer sensitivities and the informatization level of the virtual showroom. The findings underscore how virtual showrooms influence offline sales through nuanced consumer behavior dynamics. Additionally, [39] developed a heterogeneous exploration-exploitation framework for dynamic pricing in TEPC, balancing revenue maximization and demand understanding. The framework utilized weighted linear regression and the Z-Price Changes Algorithm for exploration, alongside multi-armed bandit methods like Upper Confidence Bound and Thompson Sampling for exploitation. Testing over 361 days with 10,000 demand functions revealed the Thompson Sampling Method as the most effective for maximizing revenue and market share, offering a sophisticated approach to dynamic pricing.

2.4. Strategic decisions in production capacity allocation

This category examines production capacity allocation strategies, focusing on the trade-offs between dedicated and shared production capacity approaches. These decisions significantly impact cost management, flexibility, and responsiveness to customer demand [12,33]. [28,27] and [20] analyzed the alignment of public and private objectives in restructured electricity markets. The study highlighted how private companies (PCs), focused on maximizing profits, often diverge from social welfare goals. Using an option-based approach, the research quantified welfare losses from these conflicting objectives and proposed a proactive, shared transmission-investment strategy. This strategy leverages the dependency of PCs on transmission system operator (TSO) infrastructure, significantly reducing welfare losses in uncertain market conditions. [3] investigated the strategic choice between flexible and dedicated production technologies in a two-product market under uncertain demand. Flexible technology offers adaptability, while dedicated technology provides firm production commitments, offering strategic advantages. The study revealed that while both firms often prefer dedicated technology, such a choice does not lead to equilibrium in sequential decision-making games. In highly uncertain environments, incumbents strategically overinvest in flexible technology to compel entrants toward dedicated technology, showcasing how uncertainty shapes strategic capacity decisions.

2.5. Research gap and contribution

Despite the extensive application of Stackelberg competition models in supply chain management, most existing studies focus on isolated interactions, either within a single supply chain or among competing entities, but not both simultaneously. The literature lacks an integrated perspective that considers both intra-chain dynamics (manufacturer–distributor coordination) and inter-chain rivalry (dual supply chains competing) under heterogeneous customer behaviors. Moreover, the strategic implications of combining centralized and decentralized decision-making structures, particularly under conditions of dual customer sensitivity to price and delivery time, have not been sufficiently investigated.

This study addresses these gaps by introducing a novel game-theoretic model that captures layered competition within and between two supply chains. The framework integrates dual customer class differentiation, ten mixed strategic configurations, and a nested

Stackelberg structure, allowing for the exploration of hybrid market scenarios. The contributions are both theoretical by extending the modeling of hierarchical interactions and practical by offering insights into optimal coordination and pricing strategies in competitive settings.

Table 1 provides a detailed classification of relevant research, highlighting key contributions in production capacity allocation and optimization.

3. Problem statement

This study develops a model to consider the competition between two supply chains operating in the same market, each striving to maximize profitability and market share. A clear understanding of supply chain components and their interactions is important for creating a valuable model. By using both of linear or nonlinear programming approaches, the research aims to optimize supply chain dynamics and identify strategies that enhance overall performance. The model is built on key expectations, including the presence of two supply chains, each consisting of a manufacturer and a distributor producing similar products. While these products serve the same purpose, they are differentiated based on price and delivery time. Each supply chain can operate under one of two strategic approaches: a centralized strategy, where the manufacturer and distributor collaborate to align decisions making, or a decentralized optimization strategy, where each entity makes independent options, fostering internal competition.

Additionally, the model incorporates two critical behavioral coefficients to reflect real-world market conditions. Supply chain agility is represented by the responsiveness coefficient γ, which quantifies how rapidly a supply chain can adjust its delivery times in response to customer expectations or operational shifts. Environmental uncertainty is captured using the volatility coefficient β, reflecting customer sensitivity to price changes and perceived service reliability. These coefficients are integrated into the demand functions and significantly influence the competitive outcomes under different strategy combinations. Furthermore, the sensitivity coefficient θ accounts for the customer reaction to internal disparities within the same supply chain, such as inconsistent pricing or service levels between the producer and distributor. This parameter allows the model to capture the effects of internal coordination (or lack thereof) on customer satisfaction and demand.

The market is separated into two different customer classes: low-class customers who value price and high-class customers who are extremely sensitive to delivery time. One supply chain serves each customer class exclusively, and demand is impacted by the competitor

Table 1
Comparison of parameters in various studies.

Source	Competition Variables			Coordination Strategy		Production Capacity Approach		Customer Differentiation		Solution Approach
	Price	Delivery Time	Others	Centralized	Decentralized	Shared	Dedicated	Homogeneous	Heterogenous	
Klimas, et al., 2024					✔					Deep learning
[22]					✔					Metaheuristic Algorithms
[13]	✔									Deep learning
[39]									✔	Z-Price Changes Algorithm
[44]		✔								Delayed feedback control
[35]								✔		Deep learning
[22]					✔					Metaheuristic Algorithms
[24]	✔									Metaheuristic Algorithms
[7]		✔								Bayesian learning
[20]						✔				Deep learning
[43]							✔			Metaheuristic Algorithms
[42]	✔									Sustainable supplier selection
Zhang, et al. 2022	✔									Metaheuristic Algorithms
[40]							✔			Game theory
[11]								✔		Deep learning
Chen, et al., 2020				✔						Metaheuristic Algorithms
[14]	✔									Game theoretical approach
Mahmoodi, 2020	✔	✔								Metaheuristic Algorithms
[19]							✔			Game theory
[45]				✔						Metaheuristic Algorithms
[17]	✔	✔				✔	✔		✔	Game theory
[3]							✔			Game theory
[32]	✔	✔								Metaheuristic Algorithms
[41]		✔								Metaheuristic Algorithms
[8]	✔	✔								Metaheuristic Algorithms
[16]	✔									Metaheuristic Algorithms
[12]						✔				Metaheuristic Algorithms
[33]							✔			Metaheuristic Algorithms
[48]	✔	✔								Metaheuristic Algorithms
[46]	✔	✔								Metaheuristic Algorithms
[9]	✔									Metaheuristic Algorithms
[21]					✔					Metaheuristic Algorithms
[47]								✔		Metaheuristic Algorithms
[31]				✔						Metaheuristic Algorithms
Hausman, et al., 2002						✔				Metaheuristic Algorithms
Otley, 2002				✔						Metaheuristic Algorithms
Current Research	✔	✔		✔	✔	✔	✔		✔	Game theory, Heuristic and Metaheuristic Algorithms

chain's pricing and delivery policies as well as its own. This assumption reflects real-world segmentation strategies in dual-channel systems such as e-commerce and retail electronics. Customer demand follows a Poisson distribution, a widely adopted assumption in queuing-based supply chain models due to its ability to represent random customer arrivals with analytical tractability. Service within each chain operates under a first-come, first-served (FCFS) queuing system, which realistically mirrors the operational flow of many service-based and order-driven fulfillment systems. The use of two supply chains and two customer classes provides a simplified yet strategically rich abstraction that enables comparative analysis across different competition structures. In centralized supply chains, the manufacturer and distributor jointly determine a single price and delivery time. Conversely, in decentralized supply chains, the manufacturer proposes pricing to the distributor, leading to two separate prices and a unified delivery schedule. This structured formulation supports strategic evaluation across cooperation, coordination, and competitive scenarios in dual supply chains.

4. Model development

The definitions of sets, parameters, and variables are provided in the following tables:

Sets:

$\boldsymbol{i,j}$	Supply chains $\boldsymbol{i}\epsilon\{\boldsymbol{1},\boldsymbol{2}\}$ and their counterparts $\boldsymbol{j}\epsilon\{\boldsymbol{3}-\boldsymbol{i}\}$
$\boldsymbol{k}$	Customer classes $k\epsilon\{h,l\}$

Parameters:

Sensitivity coefficients of customer demand to price and delivery time ($\beta_P^k,\ \beta_L^k$)

Sensitivity coefficients of customer demand to differences between chains (γ_p,γ_L)

Sensitivity coefficients of customer demand to differences within a chain ($\theta_p,\ \theta_L$)

Total market demand, chain infrastructure, advertising, operating costs ($a^i,\ m_m^i, m_d^i,\ A^i$)

Coordination establishment cost, chain service level ($\phi^i,\ \alpha^i$)

Customer waiting time in steady state (W_k^i)

Actual service level realized by the chain (S_k^i)

Variables:

Prices and delivery times offered by the chains (P_h^i, L_h^i)

Customer demand rates and production rates ($\lambda_k^i,\ \mu_k^i$)

Profit Function:

Maximizing the chain's profit considering revenue from sales, production, advertising, coordination, and subtracting production and operating costs.

To assess the attractiveness of products for two classes of customers, both chains are evaluated based on price, delivery time, and the disparities between them. The customer demand rate for each chain is determined through the following method:

$$\lambda_h^i = a^i - \beta_P^h P_h^i + \theta_p\left(P_l^i - P_h^i\right) - \beta_L^h L_h^i + \theta_L\left(L_l^i - L_h^i\right) + \gamma_p\left(P_h^j - P_h^i\right) + \gamma_L\left(L_h^j - L_h^i\right); i\epsilon\{1,2\},\quad j = 3-i \tag{1}$$

$$\lambda_l^i = a^i - \beta_P^l P_l^i + \theta_p\left(P_h^i - P_l^i\right) - \beta_L^l L_l^i + \theta_L\left(L_h^i - L_l^i\right) + \gamma_p\left(P_l^j - P_l^i\right) + \gamma_L\left(L_l^j - L_l^i\right); i\epsilon\{1,2\},\quad j = 3-i \tag{2}$$

Lemma 1

Consider a production system with two classes of customers, where the producer can opt for either shared or dedicated capacity policies. In the dedicated production mode, the waiting time distribution function is given by $S_k^i\left(L_k^i\right) = 1 - e^{\left(\lambda_k^i-\mu_k^i\right)L_k^i}$, where λ_k^i and μ_k^i represent the input and service rates, respectively. For shared capacity production, the waiting time distribution is given by

$$S_l^k(L_l) = 1 - \overline{S_l^k}(L_l) = \sum_{n=0}^{\infty} e^{-\gamma L_l}\frac{(\gamma L_l)^n}{n!}X_0(I-R)^{-1}H_n e \tag{3}$$

The proof of Lemma 1 is provided in appendix 1

Consider a production system with two classes of customers, where the producer can opt for either shared or dedicated capacity policies. In the dedicated production mode, the waiting time distribution function is given by $S_k^i\left(L_k^i\right) = P\left(W_k^i \le L_k^i\right) = 1 - e^{\left(\lambda_k^i-\mu_k^i\right)L_k^i} \ge \alpha$, where λ_k^i and μ_k^i represent the input and service rates, respectively. For shared capacity production, the waiting time distribution is given by

$$S_l^k(L_l) = 1 - \overline{S_l^k}(L_l) = \sum_{n=0}^{\infty} e^{-\gamma L_l}\frac{(\gamma L_l)^n}{n!}X_0(I-R)^{-1}H_n e.$$

Following the different states of the model, it is formulated for only one chain as shown in Fig. 3:

The proposed modeling framework incorporates four distinct configurations of supply chain strategy, arising from the combination of coordination structure (centralized vs. decentralized) and production capacity policy (dedicated vs. shared). These configurations include: CSDC (Centralized structure with Dedicated Capacity), CSSC (Centralized structure with Shared Capacity), DSDC (Decentralized structure with Dedicated Capacity), and DSSC (Decentralized structure with Shared Capacity), as illustrated in Fig. 3. Each configuration reflects a realistic decision environment where supply chain members choose between fully coordinated or independent decision-making and decide whether to allocate dedicated resources to high-priority customers or share capacity across all segments. The following subsection introduces the mathematical formulation of the CSDC model, which assumes full coordination between manufacturer and distributor with dedicated production capacity for each customer class.

4.1. CSDC model (centralized structure with dedicated capacity)

This model represents a fully coordinated supply chain in which the manufacturer and distributor jointly decide on pricing and service levels. The production capacity is dedicated, meaning separate resources are allocated to each customer class. This configuration aims to maximize responsiveness for high-priority customers while maintaining full internal coordination.

$$\begin{aligned}\mathrm{Max}^{\mathrm{CSDS}}\Pi^{\mathrm{i}} = & -\left(\beta_{\mathrm{p}}^{\mathrm{h}}+\theta_{\mathrm{p}}+\gamma_{\mathrm{p}}\right)\left(\mathrm{P}_{\mathrm{h}}^{\mathrm{i}}\right)^2-\left(\beta_{\mathrm{p}}^{\mathrm{l}}+\theta_{\mathrm{p}}+\gamma_{\mathrm{p}}\right)\left(\mathrm{P}_{\mathrm{l}}^{\mathrm{i}}\right)^2+2\quad\theta_{\mathrm{p}}\mathrm{P}_{\mathrm{h}}^{\mathrm{i}}\mathrm{P}_{\mathrm{l}}^{\mathrm{i}}\\ &+\left\{\mathrm{a}^{\mathrm{i}}-\beta_{\mathrm{L}}^{\mathrm{h}}\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}+\theta_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}-\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}\right)\quad+\gamma_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{h}}^{\mathrm{j}}-\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}\right)+\gamma_{\mathrm{p}}\mathrm{P}_{\mathrm{h}}^{\mathrm{j}}\right.\\ &\left.+\left(\mathrm{m}_{\mathrm{m}}^{\mathrm{i}}+\mathrm{m}_{\mathrm{d}}^{\mathrm{i}}\right)\left(\beta_{\mathrm{p}}^{\mathrm{h}}+\theta_{\mathrm{p}}+\gamma_{\mathrm{p}}\right)\right\}\mathrm{P}_{\mathrm{h}}^{\mathrm{i}}\\ &+\left\{\mathrm{a}^{\mathrm{i}}-\beta_{\mathrm{L}}^{\mathrm{l}}\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}+\theta_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}-\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}\right)\quad+\gamma_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{l}}^{\mathrm{j}}-\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}\right)+\gamma_{\mathrm{p}}\mathrm{P}_{\mathrm{l}}^{\mathrm{j}}\right.\\ &\left.+\left(\mathrm{m}_{\mathrm{m}}^{\mathrm{i}}+\mathrm{m}_{\mathrm{d}}^{\mathrm{i}}\right)\left(\beta_{\mathrm{p}}^{\mathrm{l}}+\theta_{\mathrm{p}}+\gamma_{\mathrm{p}}\right)\right\}\mathrm{P}_{\mathrm{l}}^{\mathrm{i}}-\mathrm{A}^{\mathrm{i}}\left(\mu_{\mathrm{h}}^{\mathrm{i}}+\mu_{\mathrm{l}}^{\mathrm{i}}\right)\\ &+\left\{2\mathrm{a}^{\mathrm{i}}+\left(\beta_{\mathrm{L}}^{\mathrm{h}}+\gamma_{\mathrm{L}}\right)\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}+\left(\beta_{\mathrm{L}}^{\mathrm{l}}+\gamma_{\mathrm{L}}\right)\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}-\gamma_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{h}}^{\mathrm{j}}+\mathrm{L}_{\mathrm{l}}^{\mathrm{j}}\right)\right.\\ &\left.-\gamma_{\mathrm{p}}\left(\mathrm{P}_{\mathrm{h}}^{\mathrm{j}}+\mathrm{P}_{\mathrm{l}}^{\mathrm{j}}\right)\right\}\left(\mathrm{m}_{\mathrm{m}}^{\mathrm{i}}+\mathrm{m}_{\mathrm{d}}^{\mathrm{i}}\right)-\phi^{\mathrm{i}}\end{aligned} \tag{4}$$

$$S.t:\ -\left(\beta_P^h+\theta_p+\gamma_p\right)P_h^i+\theta_p P_l^i+\theta_l L_l^i-\mu_h^i \le \frac{\ln(1-\alpha)}{L_h^i}-a^i+\left(\beta_L^h+\theta_L+\gamma_L\right)L_h^i-\gamma_p P_h^j-\gamma_L L_h^j \tag{5}$$

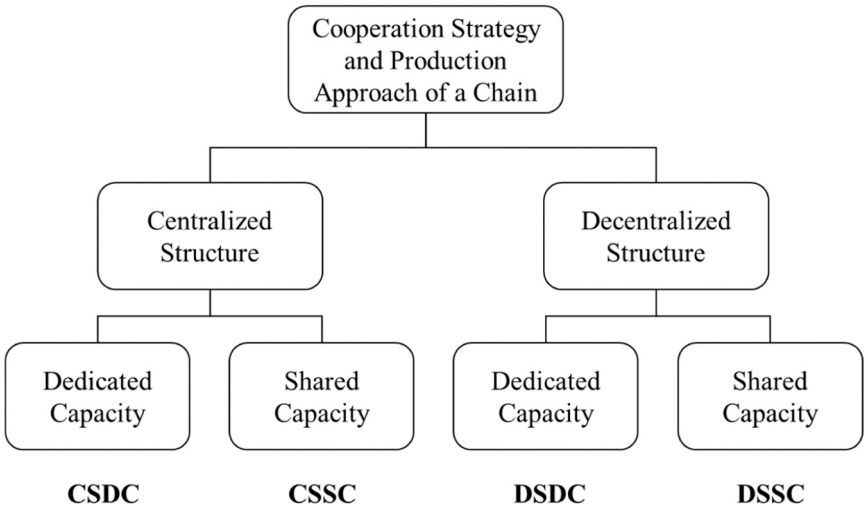


Fig. 3. Various modes of cooperation strategy and production capacity approach of a chain.

$$-\left(\beta_P^h+\theta_p+\gamma_p\right)P_l^i+\theta_pP_h^i+\theta_lL_h^i-\mu_l^i\le\frac{\ln(1-\alpha)}{L_l^i}-\alpha^i+\left(\beta_L^h+\theta_L+\gamma_L\right)L_l^i-\gamma_pP_l^j-\gamma_LL_l^j \tag{6}$$

$$-\left(\beta_P^h+\theta_p+\gamma_p\right)P_h^i+\theta_pP_l^i\ge\left(\beta_L^h+\theta_L+\gamma_L\right)L_h^i-\theta_LL_l^i-\gamma_pP_h^j-\gamma_LL_h^j-\alpha^i \tag{7}$$

$$\theta_pP_h^i-\left(\beta_P^l+\theta_p+\gamma_p\right)P_l^i\ge-\theta_LL_h^i+\left(\beta_L^l+\theta_L+\gamma_L\right)L_l^i-\gamma_pP_l^j-\gamma_LL_l^j-\alpha^i \tag{8}$$

$$L_h^i\le L_l^i \tag{8}$$

$$P_h^i,P_l^i,L_h^i,L_l^i,\mu_h^i,\mu_l^i\ge0 \tag{9}$$

The objective of this problem is to maximize the supply chain's profit, as highlighted by the objective function (4). This function accounts for the total income generated from sales while deducting the total costs associated with production, advertising, and coordination among the chain components. To achieve this goal, several key constraints are introduced to ensure operational efficiency and service quality.

First, service quality constraints (5) and (6) enforce minimum waiting time requirements for high-class and low-class customers, maintaining a predefined level of service quality. Second, service rate constraints (7) and (8) ensure that service rates remain non-negative for both customer classes, reflecting realistic operational conditions. Third, a priority constraint (9) is established to guarantee that high-class customers experience shorter waiting times than low-class customers, aligning with their sensitivity to delivery speed. Finally, the decision variable constraint (10) ensures the non-negativity of all decision variables, maintaining practical feasibility. Together, these constraints create a structured framework that balances profitability, service quality, and operational practicality while addressing the distinct needs of both customer classes.

Theorem 1. The CSDC model exhibits strict concavity with respect to the variables P_h^i,P_l^i,L_h^i,L_l^i. The proof of this theorem is provided in Appendix 2.

Therefore, the model will have a unique optimal solution, and KKT is a necessary and sufficient condition to find the solution.

With a numerical example, we solve the problem for the case where the competitor's production strategy and policy are known: $(P_h^j,P_l^j,L_h^j,L_l^j)=(10,\ 8,\ 0.5,\ 1)$.

To support the scenario analysis, this study simulates two representative supply chains operating in the consumer electronics sector. The chosen parameter values are inspired by empirical ranges and modeling assumptions drawn from previous literature (e.g., [8,32]. This configuration reflects the operations of medium-sized firms in a competitive market and provides a robust basis for scenario-based strategic evaluation.

The values of the problem parameters are according to Table 2.

The quadratic nonlinear programming model with nonlinear constraints is used to solve the problem, and the optimal values are obtained, as shown in Table 3.

It is observed that the first chain should choose a price of 8.8873 and a delivery time of 0.8454 for high-class customers and a price of 7.9838 and a delivery time of 1.0359 for low-class customers to maximize its profit in competition with the second chain.

If the delivery time for customers with low-class L_l^j of the competing chain increases, the behavior of the second chain will change. Fig. 4 shows that the demand rate for the first chain increases with an increase in the waiting time of low-class customers in the competing chain. Consequently, the first chain can offer higher prices and reduce waiting times for both classes, increasing its profit. A similar trend is observed for an increase in the waiting time of high-class customers of the competing chain.

Increasing the waiting time of low-class customers of chain j allows chain i to increase its prices and profit. However, to maintain optimal customer satisfaction, chain i must also reduce waiting times for both classes of customers, resulting in an increase in demand rate and profit for the chain. The same is valid for increasing the waiting time of high-

Table 2
A numerical example for the CSDC model.

β_p^h	β_p^l	θ_p	β_L^h	β_L^l	θ_L	γ_p	γ_L	a^1	m_m^1	m_d^1	A^1	α
0.5	0.7	0.4	0.9	0.7	0.1	0.4	0.4	10	2.9	0.1	1	0.99

Table 3
Optimal solution values for the CSDC model.

P_h^i	P_l^i	L_h^i	L_l^i	λ_h^i	λ_l^i	μ_h^i	μ_l^i	Π_{CSDC}^i
8.8873	7.9838	0.8454	1.0359	4.7600	4.0208	10.2075	8.4665	26.3885

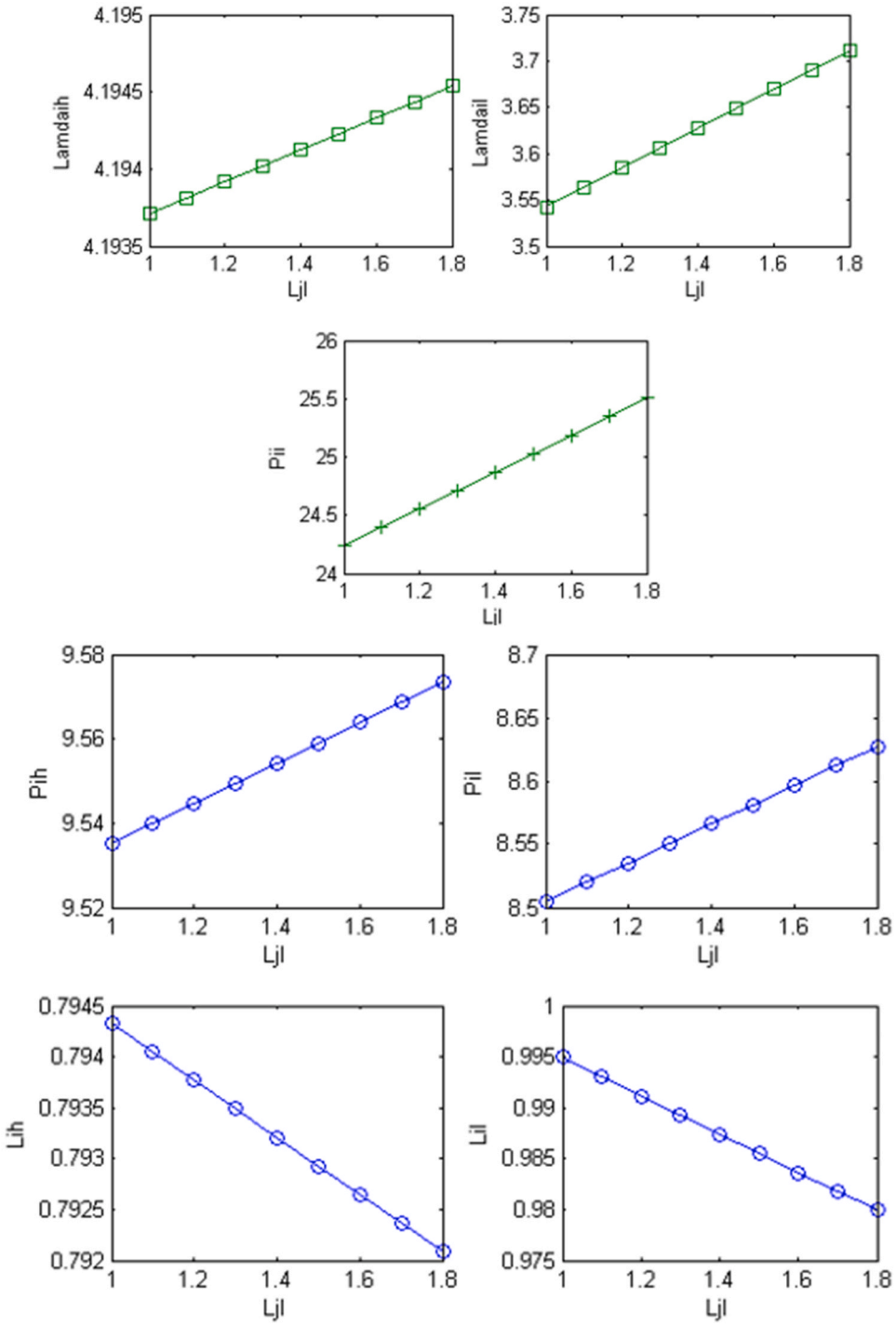


Fig. 4. Effect of increased waiting time for low-class customers (chain *j*) on chain *i* parameters.

class customers of chain j, which has a double impact on chain *i*. The trend of this increase can be seen in Fig. 5.

Increasing the waiting times for both classes of customers of the competitor chain results in an increased desire for chain *i* and an increased demand rate. Chain *i* can then increase its prices and waiting times to maximize profit. As shown in the last diagram, this results in a

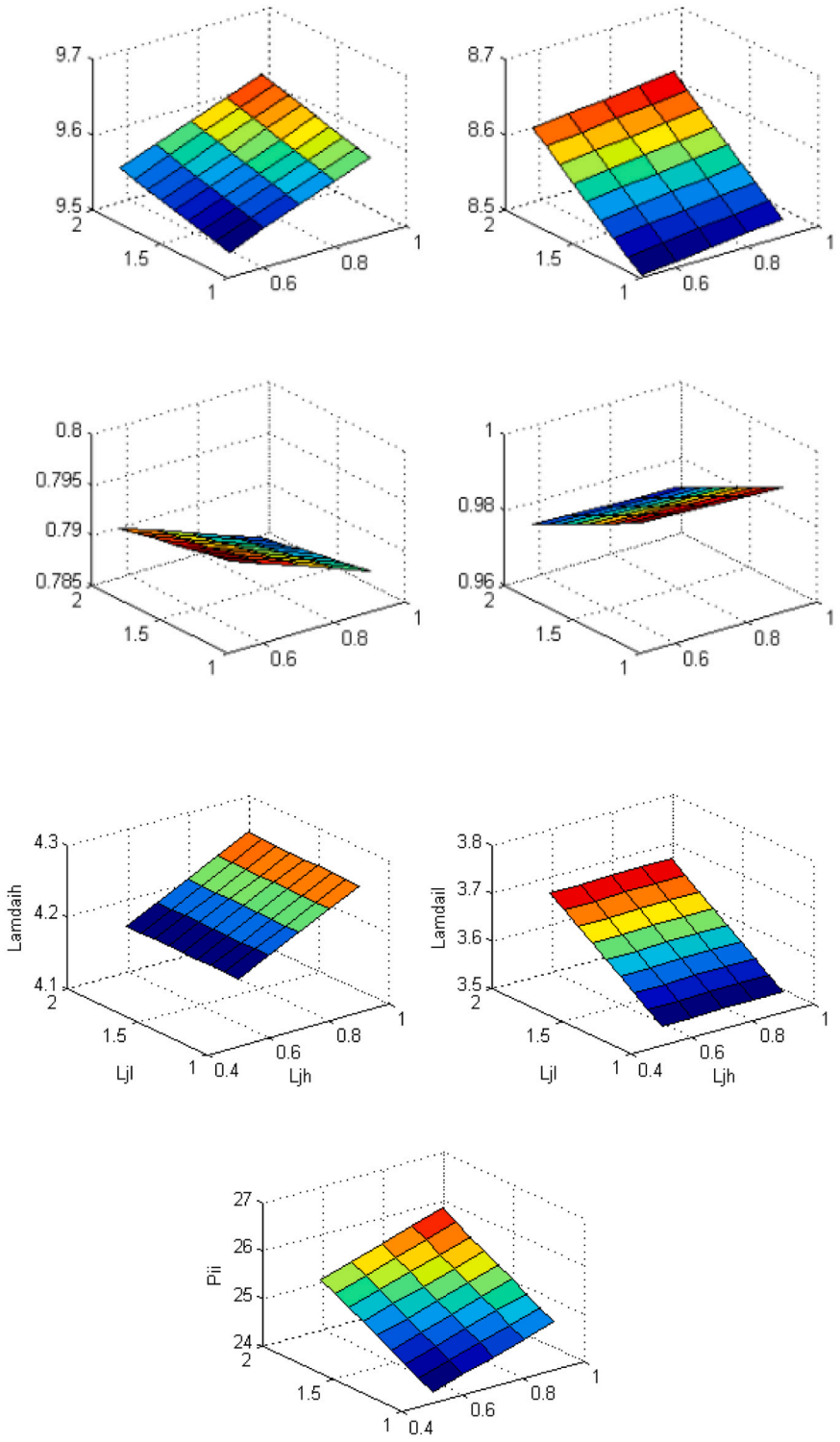


Fig. 5. Effects of increased waiting time for both customers classes (chain j) on chain i parameters.

steep increase in the chain's profits.

4.2. CSSC model (centralized structure with shared capacity)

In this configuration, the supply chain operates under a centralized structure, but the production capacity is shared across customer classes. While coordination between the manufacturer and distributor remains intact, the lack of separation in capacity allocation introduces trade-offs between service differentiation and cost efficiency.

$$\mathrm{Max}^{\mathrm{CSSC}}\Pi^{i} = -\left(\beta_{P}^{h}+\theta_{p}+\gamma_{p}\right)\left(P_{h}^{i}\right)^{2}-\left(\beta_{P}^{l}+\theta_{p}+\gamma_{p}\right)\left(P_{l}^{i}\right)^{2}+2\quad\theta_{p}P_{h}^{i}P_{l}^{i}+$$

$$\left\{a^i-\beta_L^h L_h^i+\theta_L\left(L_l^i-L_h^i\right)\quad+\gamma_L\left(L_h^j-L_h^i\right)+\gamma_p P_h^j+\left(m_m^i+m_d^i\right)\left(\beta_P^h+\theta_p+\gamma_p\right)\right\}P_h^i+\left\{a^i-\beta_L^l L_l^i+\theta_L\left(L_h^i-L_l^i\right)\quad+\gamma_L\left(L_l^j-L_l^i\right)+\gamma_p P_l^j+\left(m_m^i+m_d^i\right)\left(\beta_P^l+\gamma_p\right)\right\}P_l^i-A^i\left(\mu^i\right)+\left\{2a^i+\left(\beta_L^h+\theta_p+\gamma_L\right)L_h^i+\left(\beta_L^l+\gamma_L\right)L_l^i-\gamma_L\left(L_h^j+L_l^j\right)-\gamma_p\left(P_h^j+P_l^j\right)\right\}\left(m_m^i+m_d^i\right)-\phi^i \tag{11}$$

$$\text{S.t}: \quad -\left(\beta_P^h+\theta_p+\gamma_p\right)P_h^i+\theta_p P_l^i-\mu^i < \frac{\ln(1-\alpha)}{L_h^i}+\left(\beta_L^h+\theta_L+\gamma_L\right)L_h^i-\theta_L L_l^i-\gamma_p P_h^j-\gamma_L L_h^j-a^i \tag{12}$$

$$\sum_{n=0}^{\infty} e^{-\gamma L_l}\frac{(\gamma L_l)^n}{n!}X_0(I-R)^{-1}H_n e \geq \alpha \tag{13}$$

$$-\left(\beta_P^h+\gamma_p\right)P_h^i-\left(\beta_P^l+\gamma_p\right)P_l^i-\mu^i < \left(\beta_L^h+\gamma_L\right)L_h^i+\left(\beta_L^l+\gamma_L\right)L_l^i-\gamma_p\left(P_h^j+P_l^j\right)-\gamma_L\left(L_h^j+L_l^j\right)-2a^i \tag{14}$$

$$-\left(\beta_P^h+\theta_p+\gamma_p\right)P_h^i+\theta_p P_l^i \geq \left(\beta_L^h+\theta_L+\gamma_L\right)L_h^i-\theta_L L_l^i-\theta_p P_h^j-\gamma_L L_h^j-a^i \tag{15}$$

$$\theta_p P_h^i-\left(\beta_P^l+\theta_p+\gamma_p\right)P_l^i \geq -\theta_L L_h^i+\left(\beta_L^l+\theta_L+\gamma_L\right)L_l^i-\gamma_p P_l^j-\gamma_L L_l^j-a^i \tag{16}$$

$$L_h^i \leq L_l^i \tag{17}$$

$$P_h^i, P_l^i, L_h^i, L_l^i, \mu^i \geq 0 \tag{18}$$

The model's objective (11) is to maximize profit. The first constraint (12) specifies the waiting time of high-class customers, considering their priority over low-class customers. When high-class customers arrive, service to low-class customers is interrupted. Thus, the waiting time for high-class customers is like that in dedicated production conditions. The second constraint (13) indicates that the probability of the waiting time for low-class customers being less than a particular value is more than the defined confidence level α The third constraint (14) shows that the total demand rate is less than the service charge, $\lambda_h^i+\lambda_l^i-\mu^i<0$. The fourth and fifth constraints (15) and (16) depict the positive rate of entering customers of both classes. The fifth constraint (17) is a self-evident constraint that specifies the shorter waiting time for customers with a higher class. Finally, the sixth constraint (18) ensures that the decision variables are positive.

Theorem 2. The distribution of waiting time of low-class customers, denoted by $S_l^i(.)$, is convex for the price parameters P_h^i, P_l^i, delivery times, L_h^i, L_l^i and service rate μ^i. Proof of Theorem 2 is provided in Appendix 2.

After substituting the value of $S_l^i(.)$ at the known point (P_h^{ik}, P_l^{ik}, L_h^{ik}, L_l^{ik}, μ^{ik}) in Eq. (13), we obtain $S_l^{ik}(.)$. Substituting this in Eq. (13), we can rewrite the constraint as follows.

$$S_l^{ik}(.)+\left(P_h^i-P_h^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial P_h^i}\right)+\left(P_l^i-P_l^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial P_l^i}\right)+\left(L_h^i-L_h^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial L_h^i}\right)+\left(L_l^i-L_l^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial L_l^i}\right)+\left(\mu^i-\mu^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\mu^i}\right)>\alpha \tag{19}$$

Theorem 3. The KKT conditions are suitable for the CSSC model and

Table 4
Optimal solution values for the CSSC model (without reliability constraints).

P_h^i	P_l^i	L_h^i	L_l^i	μ^i	λ_h^i	λ_l^i
9.0507	7.6836	0.6269	0.6269	12.038	4.6925	5.0052

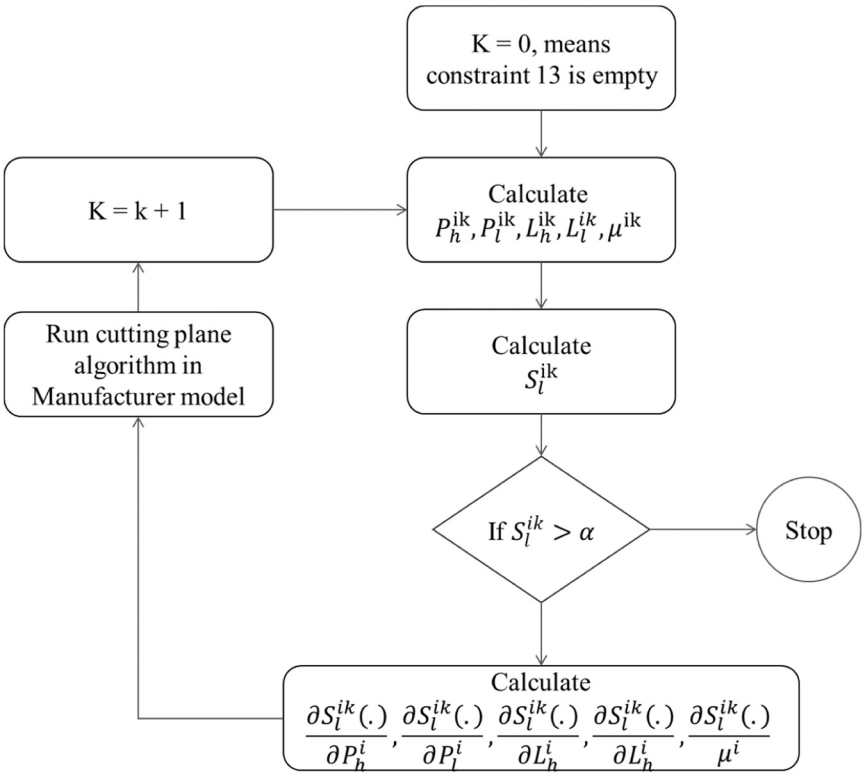


Fig. 6. Algorithm for solving the CSSC Model.

Table 5
Parameters value for the last five iterations of the CSSC model.

No. Repetition	*Parameter values* (P_h^{ik}, P_l^{ik}, L_h^{ik}, L_l^{ik}, μ^{ik})	*Confidence level values* (S_k^i, S_l^i)	*Cuts created* $\left(\frac{\partial S_l^{ik}(.)}{\partial P_h^i}\right), \left(\frac{\partial S_l^{ik}(.)}{\partial P_l^i}\right), \left(\frac{\partial S_l^{ik}(.)}{\partial L_h^i}\right), \left(\frac{\partial S_l^{ik}(.)}{\partial L_l^i}\right)$
11	(9.14, 7.81, 0.55, 0.94, 13.12)	(0.99,0.8577)	(0.08, 0.05, 0.11,0.28,0.06)
12	(9.32, 8.12, 0.38, 0.38, 17.02)	(0.99,0.8895)	(0.06, 0.03, 0.07,0.66,0.04)
13	(9.03, 7.67, 0.71, 0.93, 11.1)	(0.99,0.9387)	(0.23, 0.19, 0.31,0.55,0.2)
14	(9.16, 7.92, 0.51, 0.51, 13.87)	(0.99,0.9594)	(0.1, 0.07, 0.14,0.65,0.08)
15	(9.21, 7.93, 0.48, 0.79, 14.4)	(0.99,0.9968)	Stopping the algorithm

are sufficient and necessary for finding the optimal solution to the problem. Proof of Theorem 3 is available in Appendix 2.

We use the algorithm of cutting plane methods to solve the above problem. The framework of this algorithm is like the method proposed by (Kelley, 1960). The main difference from previous algorithms is that the GP method is used here to generate cuts: the general algorithm is to first solve the problem as a QP without considering constraint (13) and obtain an initial value (P_h^0, P_l^0,L_h^0, L_l^0, μ^0). Then we use GP to obtain the distribution $S_l^{i(P_h^0,\ P_l^0,L_h^0,\ L_l^0\ \ \mu^0)}(.)$. If the value of $S_l^{i(P_h^0,\ P_l^0,L_h^0,\ L_l^0\ \ \mu^0)}(.)$ satisfies the confidence level considered for serving customers (α), we stop solving the problem. Otherwise, we put the obtained initial value in Eq. (19), a linear equation, and solve the problem again. As demonstrated in Fig. 6, the algorithm for solving the problem for CSSC incorporates a novel approach that significantly enhances the efficiency and accuracy of collaborative signal and sensor control systems.

The algorithm continues until the reliability constraint of the chain is met. Subsequently, we will delve into the specifics of the solution method through a numerical example (such as the one presented in Table 2).

We solved the CSSC model without considering the reliability constraint (13). The corresponding results are displayed in Table 4.

Using these values, we compute the value of the function S_l^i to assess the reliability level of the supply chain. In this case, the value of S_l^i is 0.6124, less than the supply chain's committed value (0.99). Therefore, we continue the algorithm to obtain better results. As illustrated in Table 5 the algorithm stops at iteration 15, where the stopping criterion is met, i.e., the reliability level.

Next, we solve the problem for different scenarios with and without product substitution and other operational costs. The results are provided in Table 6.

4.3. DSDC model (decentralized structure with dedicated capacity)

This model assumes decentralized decision-making, where the manufacturer and distributor act independently. However, production capacity is still dedicated to customer classes, allowing some service segmentation. The absence of coordination, combined with resource separation, can lead to suboptimal decisions from a global perspective.

Table 6
Optimal solution values for the CSSC model.

	No replacement		*by replacing*			
	($\theta_P = \theta_L = 0$)		($\theta_P = 0.2$, $\theta_L = 0.5$)		($\theta_P = 0.4$, $\theta_L = 0.3$)	
	$A^i = 0.1$	$A^i = 1.0$	$A^i = 0.1$	$A^i = 1.0$	$A^i = 0.1$	$A^i = 1.0$
P_h^i	9.34	9.51	9.11	9.30	8.92	9.22
P_l^i	7.57	7.37	7.82	7.52	7.91	7.92
L_h^i	0.17	0.62	0.14	0.62	0.18	0.48
L_l^i	0.26	0.99	0.19	1.00	0.27	0.78
μ^i	32.26	12.27	39.28	12.34	31.71	14.51

$$\begin{aligned}\mathrm{Max}_{\mathrm{Mnf}}^{\mathrm{DSDC}}\Pi^{\mathrm{i}} = {} & \left(\mathrm{P}_{\mathrm{mh}}^{\mathrm{i}} - \mathrm{m}_{\mathrm{m}}^{\mathrm{i}} - \mathrm{A}^{\mathrm{i}}\right)\left\{\mathrm{a}^{\mathrm{i}} - \beta_{\mathrm{P}}^{\mathrm{h}}\mathrm{P}_{\mathrm{dh}}^{\mathrm{i}} + \theta_{\mathrm{p}}\left(\mathrm{P}_{\mathrm{dl}}^{\mathrm{i}} - \mathrm{P}_{\mathrm{dh}}^{\mathrm{i}}\right) - \beta_{\mathrm{L}}^{\mathrm{h}}\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}\right. \\ & \left. + \theta_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{l}}^{\mathrm{i}} - \mathrm{L}_{\mathrm{h}}^{\mathrm{i}}\right) + \gamma_{\mathrm{p}}\left(\mathrm{P}_{(.\mathrm{h})}^{\mathrm{j}} - \mathrm{P}_{\mathrm{dh}}^{\mathrm{i}}\right) + \gamma_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{h}}^{\mathrm{j}} - \mathrm{L}_{\mathrm{h}}^{\mathrm{i}}\right)\right\} \\ & + \left(\mathrm{P}_{\mathrm{ml}}^{\mathrm{i}} - \mathrm{m}_{\mathrm{m}}^{\mathrm{i}} - \mathrm{A}^{\mathrm{i}}\right)\left\{\mathrm{a}^{\mathrm{i}} - \beta_{\mathrm{P}}^{\mathrm{l}}\mathrm{P}_{\mathrm{dl}}^{\mathrm{i}} + \theta_{\mathrm{p}}\left(\mathrm{P}_{\mathrm{dh}}^{\mathrm{i}} - \mathrm{P}_{\mathrm{dl}}^{\mathrm{i}}\right)\right. \\ & \left. - \beta_{\mathrm{L}}^{\mathrm{l}}\mathrm{L}_{\mathrm{l}}^{\mathrm{i}} + \theta_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{h}}^{\mathrm{i}} - \mathrm{L}_{\mathrm{l}}^{\mathrm{i}}\right) + \gamma_{\mathrm{p}}\left(\mathrm{P}_{(.\mathrm{l})}^{\mathrm{j}} - \mathrm{P}_{\mathrm{dl}}^{\mathrm{i}}\right) + \gamma_{\mathrm{L}}\left(\mathrm{L}_{\mathrm{l}}^{\mathrm{j}} - \mathrm{L}_{\mathrm{l}}^{\mathrm{i}}\right)\right\} \\ & + \mathrm{A}^{\mathrm{i}}\left(\frac{\ln(1-\alpha)}{\mathrm{L}_{\mathrm{h}}^{\mathrm{i}}} + \frac{\ln(1-\alpha)}{\mathrm{L}_{\mathrm{l}}^{\mathrm{i}}}\right)\end{aligned} \tag{20}$$

$$S.t: -\left(\beta_P^h + \theta_p + \gamma_p\right)P_{dh}^i + \theta_p P_{dl}^i - \mu_h^i \le \frac{\ln(1-\alpha)}{L_h^i} - a^i + \left(\beta_L^h + \theta_L + \gamma_L\right)L_h^i - \gamma_p P_{dh}^j - \gamma_L L_h^j \tag{21}$$

$$-\left(\beta_P^h + \theta_p + \gamma_p\right)P_{dl}^i + \theta_p P_{dh}^i - \mu_l^i \le \frac{\ln(1-\alpha)}{L_l^i} - a^i + \left(\beta_L^h + \theta_L + \gamma_L\right)L_l^i - \gamma_p P_{dl}^j - \gamma_L L_l^j \tag{22}$$

$$L_h^i \le L_l^i \tag{23}$$

$$P_{mh}^i, P_{ml}^i, L_h^i, L_l^i \ge 0 \tag{24}$$

$$\begin{aligned}Max_{Dst}^{DSDC}\Pi^i = {} & \left(P_{dh}^i - P_{mh}^i - m_d^i\right)\left(a^i - \beta_P^h P_{dh}^i + \theta_p\left(P_{dl}^i - P_{dh}^i\right) - \beta_L^h L_h^i\right. \\ & \left. + \theta_L\left(L_l^i - L_h^i\right) + \gamma_p\left(P_{(.h)}^j - P_{dh}^i\right) + \gamma_L\left(L_h^j - L_h^i\right)\right) \\ & + \left(P_{dl}^i - P_{ml}^i - m_l^i\right)\left(a^i - \beta_P^l P_{dl}^i + \theta_p\left(P_{dh}^i - P_{dl}^i\right)\right. \\ & \left. - \beta_L^l L_l^i + \theta_L\left(L_h^i - L_l^i\right) + \gamma_p\left(P_{(.l)}^j - P_{dl}^i\right) + \gamma_L\left(L_l^j - L_l^i\right)\right)\end{aligned} \tag{25}$$

$$-\left(\beta_P^h + \theta_p + \gamma_p\right)P_{dh}^i + \theta_p P_{dl}^i \ge \left(\beta_L^h + \theta_L + \gamma_L\right)L_h^i - \theta_L L_l^i - \gamma_p P_{(.h)}^j - \gamma_L L_h^j - a^i \tag{26}$$

$$\theta_p P_{dh}^i - \left(\beta_P^l + \theta_p + \gamma_p\right)P_{dl}^i \ge -\theta_L L_h^i + \left(\beta_L^l + \theta_L + \gamma_L\right)L_l^i - \gamma_p P_{(.l)}^j - \gamma_L L_l^j - a^i \tag{27}$$

$$P_{dh}^i, P_{dl}^i \ge 0 \tag{28}$$

The first objective function (20) represents the producer's profit, incorporating $P_{(.h)}^j$ and $P_{(.l)}^j$ account for the uncertainty of the opposing chain's strategy, whether cooperative or competitive.

Key constraints for the producer's function include:

1. Confidence Level Constraints:
   - o Constraints (21) and (22) ensure that the probability of customers receiving service from both classes exceeds the required confidence level.
2. Priority Constraint:
   - o Constraint (23) ensures that higher-class customers experience shorter waiting times than lower-class customers.
3. Non-Negativity Constraint:

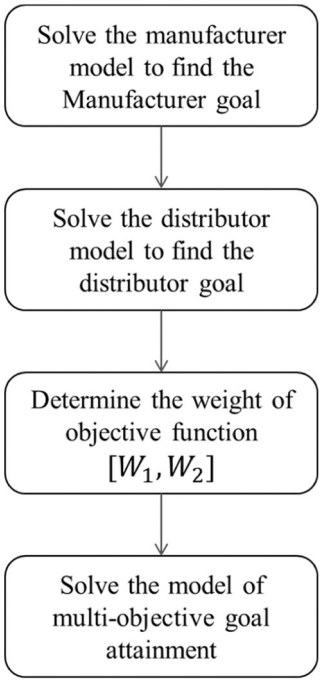


Fig. 7. Algorithm for solving the DSDC model.

- Constraint (24) guarantees that all decision variables remain non-negative.

The distributor's objective function (25) focuses on maximizing its profit independently.

Key constraints for the distributor's function include:

1. Demand Rate Constraints:
   - Constraints (26) and (27) enforce the non-negativity of customer demand rates for both classes.
2. Non-Negativity Constraint:
   - Constraint (28) ensures all decision variables are non-negative.

This dual-objective framework allows the producer and distributor to optimize their respective profits while meeting operational and service-level requirements.

Theorem 4. The DSDC model, which involves a lack of concentration of chain components and the entire production, is under KKT conditions. These conditions are necessary and sufficient to determine the solution to the problem. Proof of the theorem is provided in Appendix 2.

To illustrate the method of solving the problem, we use the same numerical example presented in the first scenario. This problem involves two objective functions: one non-linear and the other linear, along with a combination of linear and non-linear constraints. To address this complexity, we employ a multi-objective goal attainment method for its solution. As shown in Fig. 7 we begin by solving the models independently to obtain the goals. Then, we can determine the following matters by placing these target values in the model and assigning equal weight to the models.

The numerical example was such as Table 2 and the results indicate that the chain's profit is lower when there is a decentralization strategy compared to the case where the chain components cooperate. Table 7 presents the optimal values for a chain with DSDC, offering a comprehensive overview of the performance benchmarks critical to understanding and improving chain dynamics in distributed control systems.

As can be seen, in Figs. 8–11 the manufacturer's profit is highly sensitive to changes in the cost of infrastructure. As the cost of infrastructure increases, the manufacturer's profit decreases and can even become harmful.

Table 7
Optimal solution values for the DSDC model.

P^i_{mh}	P^i_{ml}	P^i_{dh}	P^i_{dl}	L^i_h	L^i_l	λ^i_h	λ^i_l	μ^i_h	μ^i_l	Π^{Mnfi}_{DC-D}	Π^{Dsti}_{DC-D}	Π^i_{DC-D}
8.84	7.94	11.36	10.01	0.84	1.03	2.36	1.98	10.28	8.56	5.17	9.62	14.79

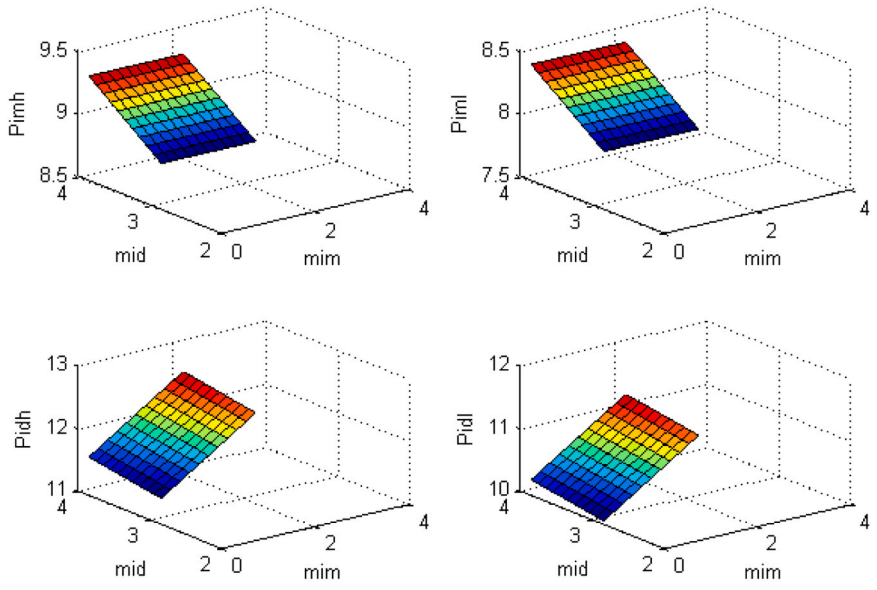


Fig. 8. Sensitivity of producer and distributor prices to infrastructure and advertising cost in DSDC model.

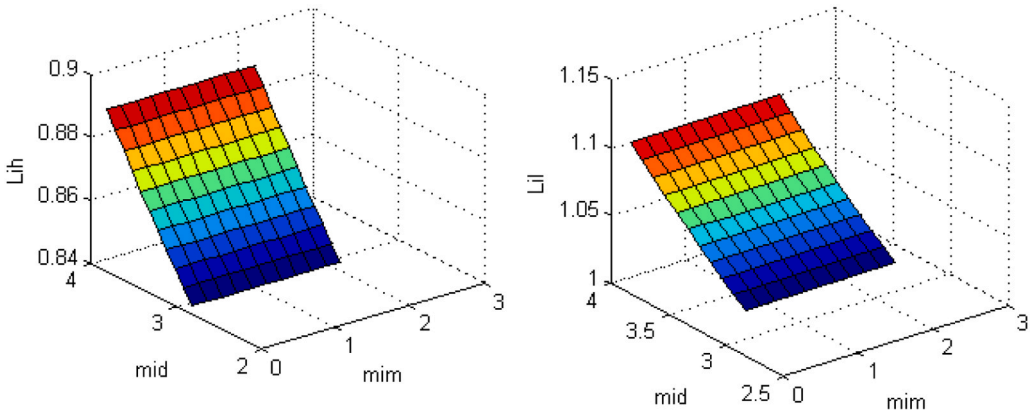


Fig. 9. Sensitivity of customer waiting times to infrastructure and advertising cost in DSDC model.

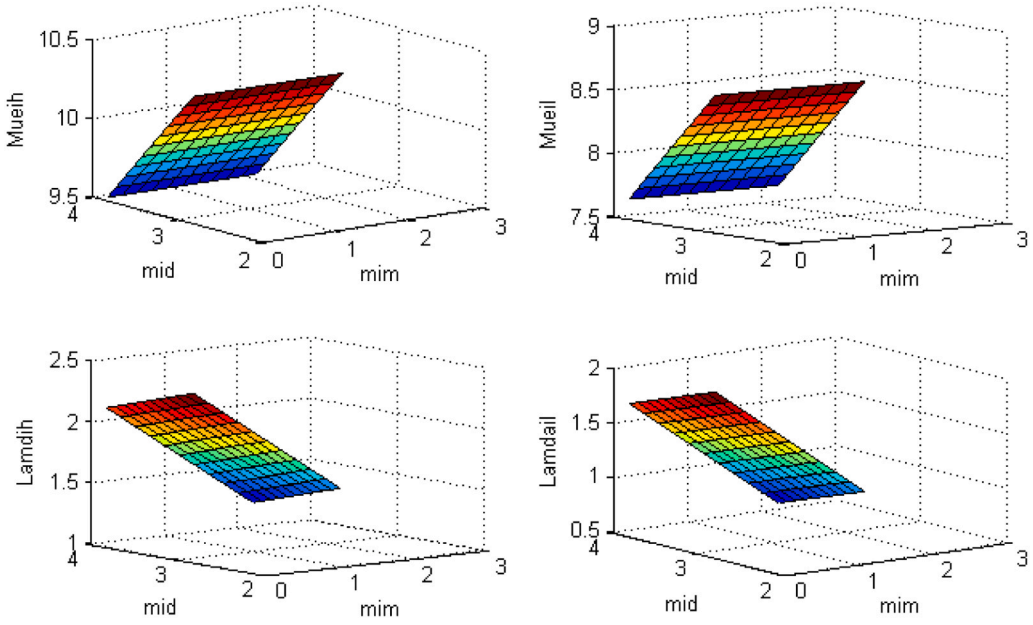


Fig. 10. Sensitivity of arrival and service rates to infrastructure and advertising cost in DSDC model.

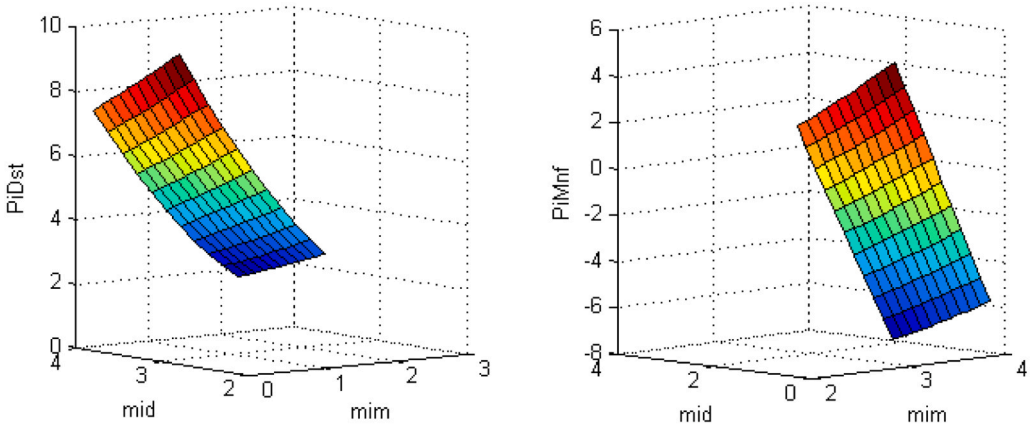


Fig. 11. Sensitivity of producer and distributor profit to infrastructure and advertising cost in DSDC model.

4.4. DSSC model (decentralized structure with shared capacity)

The DSSC model reflects a fully decentralized supply chain with shared capacity across customer segments. This configuration captures highly competitive environments with minimal coordination and limited-service differentiation, typically prioritizing cost over responsiveness.

$$\begin{aligned}\mathrm{Max}_{\mathrm{Mnf}}^{\mathrm{DSSC}}\Pi^{i} = {} & \left(P_{mh}^{i} - m_{m}^{i}\right)\left\{a^{i} - \beta_{P}^{h}P_{dh}^{i} + \theta_{p}\left(P_{dl}^{i} - P_{dh}^{i}\right) - \beta_{L}^{h}L_{h}^{i} + \theta_{L}\left(L_{l}^{i} - L_{h}^{i}\right)\right. \\ & \left. + \gamma_{p}\left(P_{(.h)}^{j} - P_{dh}^{i}\right) + \gamma_{L}\left(L_{(.h)}^{j} - L_{h}^{i}\right)\right\} \\ & + \left(P_{ml}^{i} - m_{l}^{i}\right)\left\{a^{i} - \beta_{P}^{l}P_{dl}^{i} + \theta_{p}\left(P_{dh}^{i} - P_{dl}^{i}\right) - \beta_{L}^{l}L_{l}^{i} + \theta_{L}\left(L_{h}^{i} - L_{l}^{i}\right)\right. \\ & \left. + \gamma_{p}\left(P_{(.l)}^{j} - P_{dl}^{i}\right) + \gamma_{L}\left(L_{l}^{j} - L_{l}^{i}\right)\right\} - A^{i}\mu^{i}\end{aligned} \tag{29}$$

$$\begin{aligned}\text{S.t}: \quad & \left(\beta_{P}^{h} - \theta_{p} - \gamma_{p}\right)P_{mh}^{i} + \theta_{p}P_{ml}^{i} - \mu^{i} \\ & < \frac{\ln(1-\alpha)}{L_{h}^{i}} + \left(\beta_{L}^{h} + \theta_{L} + \gamma_{L}\right)L_{h}^{i} - \theta_{L}L_{l}^{i} - \gamma_{p}P_{(.h)}^{j} - \gamma_{L}L_{h}^{j} - \alpha^{i}\end{aligned} \tag{30}$$

$$\sum_{n=0}^{\infty} e^{-\gamma L_{l}^{i}}\frac{\left(\gamma L_{l}^{i}\right)^{n}}{n!}X_{0}(I-R)^{-1}H_{n}e \geq \alpha \tag{31}$$

$$\begin{aligned}& -\left(\beta_{P}^{h} + \gamma_{p}\right)P_{mh}^{i} - \left(\beta_{P}^{l} + \gamma_{p}\right)P_{ml}^{i} - \mu^{i} < \left(\beta_{L}^{h} + \gamma_{L}\right)L_{h}^{i} + \left(\beta_{L}^{l} + \gamma_{L}\right)L_{l}^{i} \\ & \quad - \gamma_{p}\left(P_{(.h)}^{j} + P_{l}^{j}\right) - \gamma_{L}\left(L_{h}^{j} + L_{l}^{j}\right) - 2\alpha^{i}\end{aligned} \tag{32}$$

$$L_{h}^{i} \leq L_{l}^{i} \tag{33}$$

$$P_{mh}^{i}, P_{ml}^{i}, L_{h}^{i}, L_{l}^{i} \geq 0 \tag{34}$$

$$\begin{aligned}Max_{Dst}^{DSSC}\Pi^{i} = {} & \left(P_{dh}^{i} - P_{mh}^{i} - m_{d}^{i}\right)\left\{a^{i} - \beta_{P}^{h}P_{dh}^{i} + \theta_{p}\left(P_{dl}^{i} - P_{dh}^{i}\right) - \beta_{L}^{h}L_{h}^{i}\right. \\ & \left. + \theta_{L}\left(L_{l}^{i} - L_{h}^{i}\right) + \gamma_{p}\left(P_{(.h)}^{j} - P_{dh}^{i}\right) + \gamma_{L}\left(L_{h}^{j} - L_{h}^{i}\right)\right\} \\ & + \left(P_{dh}^{i} - P_{ml}^{i} - m_{l}^{i}\right)\left\{a^{i} - \beta_{P}^{l}P_{dl}^{i} + \theta_{p}\left(P_{dh}^{i} - P_{dl}^{i}\right) - \beta_{L}^{l}L_{l}^{i} + \theta_{L}\left(L_{h}^{i} - L_{l}^{i}\right)\right. \\ & \left. + \gamma_{p}\left(P_{(.l)}^{j} - P_{dl}^{i}\right) + \gamma_{L}\left(L_{l}^{j} - L_{l}^{i}\right)\right\}\end{aligned} \tag{35}$$

$$-\left(\beta_{P}^{h} + \theta_{p} + \gamma_{p}\right)P_{dh}^{i} + \theta_{p}P_{dl}^{i} \geq \left(\beta_{L}^{h} + \theta_{L} + \gamma_{L}\right)L_{h}^{i} - \theta_{L}L_{l}^{i} - \gamma_{p}P_{(.h)}^{j} - \gamma_{L}L_{h}^{j} - \alpha^{i} \tag{36}$$

$$\theta_{p}P_{dh}^{i} - \left(\beta_{P}^{l} + \theta_{p} + \gamma_{p}\right)P_{dl}^{i} \geq -\theta_{L}L_{h}^{i} + \left(\beta_{L}^{l} + \theta_{L} + \gamma_{L}\right)L_{l}^{i} - \gamma_{p}P_{(.l)}^{j} - \gamma_{L}L_{l}^{j} - \alpha^{i} \tag{37}$$

$$P_{dh}^{i}, P_{dl}^{i} \geq 0 \tag{38}$$

This model seeks to optimize the competition strategy in which the producer and distributor pursue independent profit maximization objectives.

The producer's objective function is expressed in Eq. (29), with the following constraints:

1. Waiting Time Probabilities:
   - o Constraint (30) ensures service prioritization for high-class customers, guaranteeing their service even during the service of lower-class customers.
   - o Constraint (31) defines the waiting time distribution for lower-class customers.
2. Arrival Rate Limitation:
   - o Constraint (32) limits customer arrival rates to prevent infinite queues.
3. Priority Constraint:
   - o Constraint (33) ensures that high-class customers experience shorter waiting times than low-class customers.
4. Non-Negativity:
   - o Constraint (34) ensures non-negativity of the producer's decision variables.

The distributor's profit maximization is represented by the objective function in Eq. (35), with the following constraints:

1. Non-Negativity of Arrival Rates:
   - o Constraints (36) and (37) enforce non-negative customer arrival rates for both high- and low-class customers.
2. Non-Negativity:
   - o Constraint (38) ensures non-negativity of the distributor's decision variables.

This framework balances independent profit goals for the producer and distributor while maintaining operational efficiency and prioritizing customer service quality.

Theorem 5. The DSSC model has a unique solution for the parameters $P_{mh}^{i}, P_{ml}^{i}, L_{h}^{i}, L_{l}^{i}, \quad P_{dh}^{i}, P_{dl}^{i}, \quad \mu^{i}$ and the KKT conditions are necessary and sufficient for finding the solution. Proof of the Theorem 5 is available in appendix 2.

The complexity of the problem increases due to the combination of joint production and the waiting time distribution function for low-class customers, making it difficult to solve. To address this, a heuristic and meta-heuristic algorithm, specifically the multi-objective genetic algorithm (GA Multi-Objective Goal), was used to solve the integrated models. The iterative algorithm used to obtain the target values of the producer and distributor functions is like the algorithm used in CSSC model (Fig. 6) and is shown in Fig. 12:

Fig. 12 shows that the problem is divided into two stages. The producer and distributor models are solved independently in the first stage using an innovative algorithm. This algorithm gradually improves the restriction related to the distribution function of customer expectations with the lower class. In the second stage, the target models of both problems are solved using a multi-objective genetic algorithm.

5. Results and discussion

5.1. Sensitivity analysis of behavioral coefficients

The behavioral sensitivity coefficients β (price sensitivity), γ (delivery responsiveness), and θ (internal coordination intensity) were calibrated within the range [0.1–1.0]. These values were selected based on parameter estimates from studies in high-speed logistics and customer behavior in fast-delivery retail markets (e.g., [32]. To evaluate the robustness of the model to variations in these coefficients, we applied a parameter sweep method, adjusting each value while monitoring effects on key outcomes such as prices, delivery times, customer allocations, and profits. The results, presented in Figs. 8 to 11, confirm that the model remains stable under a wide range of behavioral assumptions, reinforcing the validity and applicability of the framework. The sensitivity analysis of the model to changes in the producer's infrastructure costs and distributor's advertising is presented below.

To illustrate the solution method, we provided a numerical example such as Table 2. First, we solve the producer and distributor problems separately, and the results are in Table 8.

The values obtained from the previous calculations are then used to evaluate the confidence level of the chain using the S_{l}^{i} function. The resulting value of S_{l}^{i} is 0.9460, which is lower than the chain's committed value of 0.99. The algorithm is iteratively solved to obtain better results. In the fourth iteration, the algorithm reaches the stopping

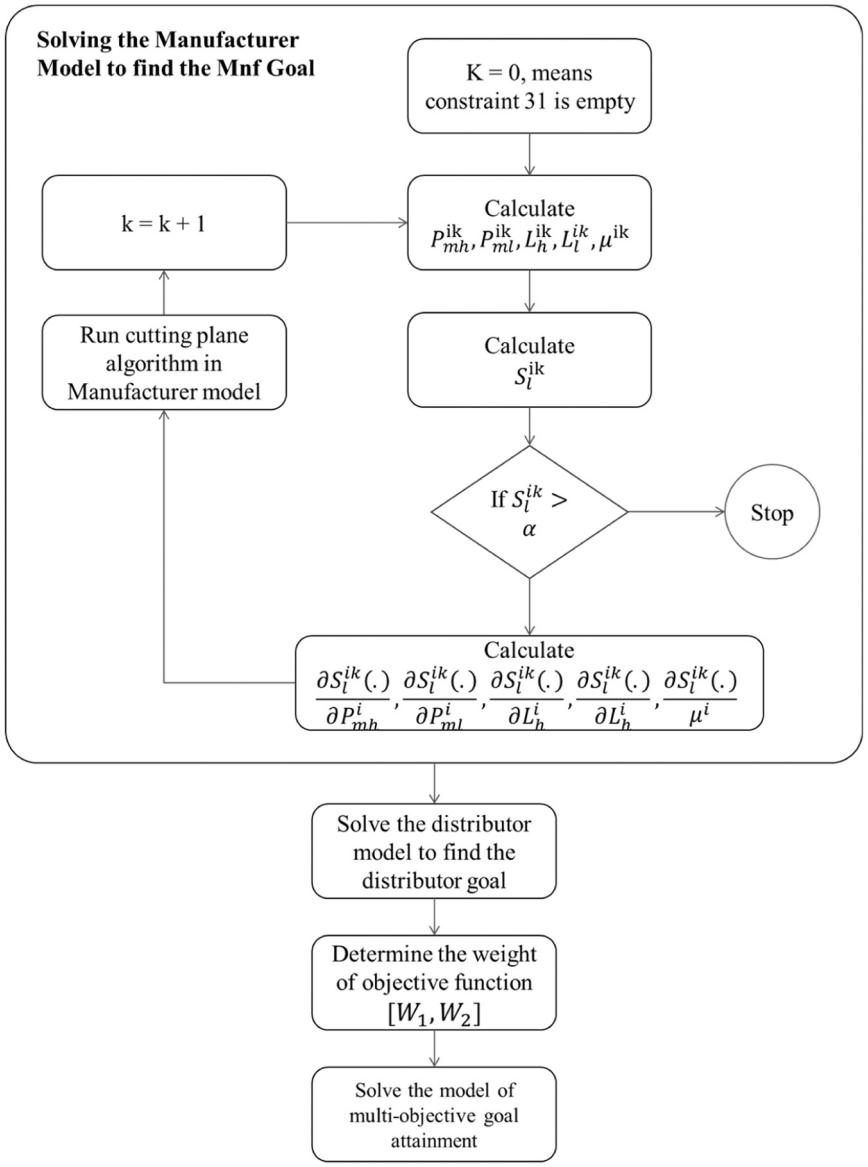


Fig. 12. Algorithm for solving the DSSC model.

condition, the confidence level, and stops. Table 9 displays the values obtained during the last five iterations.

The target values of the producer and distributor functions can be combined into a single multi-objective function, with both objectives being given equal weights. The optimal values for producer and distributor functions that can be obtained are indicated in Table 10 by solving this combined problem.

5.2. Scenario analysis

Various competition scenarios between the two chains have been examined to identify and analyze different competition modes. As shown in Table 11, ten different modes of competition arise from the combination of various modes of chain competition. The models for each chain have been developed independently and solved in the previous sections. Now we focus on finding a solution that leads to market equilibrium and stabilizes customers' prices and waiting times.

The ten strategic scenarios analyzed in this study represent an exhaustive enumeration of the decision space defined by the model. Specifically, they arise from a full-factorial combination of two structural strategies centralized and decentralized coordination and two production strategies shared vs. dedicated across the two competing supply chains. By permutating these decision types, all meaningful competitive configurations are covered, ensuring that the scenario set is both comprehensive and representative of real-world coordination possibilities.

The table shows that the chains are similar, and repeated cases have been excluded. It is assumed that the chains have complete information about each other's infrastructure costs, operational costs, reliability levels, and the number of customers. Therefore, the game used here is an

Table 8
Optimal solution values for the DSSC model (without reliability constraints).

P_{mh}^i	P_{ml}^i	P_{dh}^i	P_{dl}^i	L_h^i	L_l^i	μ^i	λ_h^i	λ_l^i
4.3348	2.5614	7.5207	6.5700	0.3878	0.5127	21.1708	7.0529	6.7465

Table 9
Parameter values for the last five iterations of the DSSC model.

No. Repetition	Parameter values (P^{ik}_{mh}, P^{ik}_{ml}, P^{ik}_{dh}, P^{ik}_{dl}, L^{ik}_{h}, L^{ik}_{l})	Confidence level values (S^{i}_{k}, S^{i}_{l})	Cuts created $\left(\frac{\partial S^{ik}_{l}(.)}{\partial P^{i}_{h}}\right), \left(\frac{\partial S^{ik}_{l}(.)}{\partial P^{i}_{l}}\right), \left(\frac{\partial S^{ik}_{l}(.)}{\partial L^{i}_{h}}\right), \left(\frac{\partial S^{ik}_{l}(.)}{\partial L^{i}_{l}}\right)$
1	(4.33, 2.56, 7.52, 6.57, 0.38, 0.51)	(0.99,0.9460)	(0.08, 0.05, 0.11,0.28,0.06)
2	(6.36, 7.25, 6.42, 0.38, 0.62)	(0.99,0.9672)	(0.06, 0.03, 0.07,0.66,0.04)
3	(4.42, 2.45, 7.33, 6.38, 0.38, 0.65)	(0.99, 0.9725)	(0.23, 0.19, 0.31,0.55,0.2)
4	(4.57, 2.63, 7.43, 6.12, 0.38, 0.69)	(0.99, 0.9992)	Stopping the algorithm

Table 10
Optimum solution values of the DSSC model.

P^{i}_{mh}	P^{i}_{ml}	P^{i}_{dh}	P^{i}_{dl}	L^{i}_{h}	L^{i}_{l}	λ^{i}_{h}	λ^{i}_{l}	μ^{i}	Π^{Mnfi}_{DSSC}	Π^{Dsti}_{DSSC}	Π^{i}_{DSSC}
7.96	7.15	10.22	9.01	0.76	0.93	2.12	1.78	9.25	4.65	8.66	13.31

Table 11
Scenarios of two chains competition in the presence of two classes of customers.

Scenario	The first chain	The second chain
1	CSDC	CSDC
2	CSDC	CSSC
3	CSDC	DSDC
4	CSDC	DSSC
5	CSSC	CSSC
6	CSSC	DSDC
7	CSSC	DSSC
8	DSDC	DSDC
9	DSDC	DSSC
10	DSSC	DSSC

Table 12
A numerical example for all scenarios of two chains competition.

β^{h}_{p}	β^{l}_{p}	θ_{p}	β^{h}_{L}	B^{l}_{L}	θ_{L}	γ_{p}	γ_{L}	α
0.5	*0.7*	*0.4*	*0.9*	*0.7*	*0.1*	*0.4*	*0.4*	*0.99*
a^1	m^1_m	m^1_d	A^1	a^2	m^2_m	m^2_d	A^2	α
10	*2.9*	*0.1*	*1.5*	*10*	*2.6*	*0.2*	*1*	*0.99*

Table 13
Optimum solution values in the competition between CSDC-CSDC.

P^{i}_{h}	P^{i}_{l}	L^{i}_{h}	L^{i}_{l}	μ^{i}_{h}	μ^{i}_{l}	λ^{i}_{h}	λ^{i}_{l}	Π^{i}_{CSDC}
11.26	10.49	0.88	1.03	11.61	10.76	6.39	6.29	63.33
P^{j}_{h}	P^{j}_{l}	L^{j}_{h}	L^{j}_{l}	μ^{j}_{h}	μ^{j}_{l}	λ^{j}_{h}	λ^{j}_{l}	Π^{j}_{CSDC}
11.14	10.23	0.89	1.05	11.51	10.31	6.34	5.94	58.79

evolutionary Stackelberg game. In this game, one of the chains acts as a leader, acts, and then responds to the competitor chain by updating its strategy, production policy, and market decisions. This process continues until the market reaches equilibrium conditions. It is also assumed that the characteristics and preferences of customers do not change during the competition.

A numerical example based on the data in Table 12:

The goal is to stop the competition between the chains, achieved through two conditions: repeating the process 1000 times, or stabilizing the parameters to a different value of 1.e-8.

In this scenario, the chain competition reaches a state of equilibrium after six stages, and the outcomes are presented in Table 13:

The stopping condition was met after 6 repetitions.

As shown in Fig. 13 price, delivery time, service rates, arrival rate for customers with a high and low class and profit of each chain (CSDC-CSDC) was illustrated.

It shows that despite having the same strategic structure and adopting similar policies, the first chain performed better during the competition due to its superior infrastructure and more effective product cost management.

5.3. Baseline comparison and strategic insights

This subsection presents a comparative evaluation of the ten competition scenarios to identify strategic insights and benchmark outcomes. Among these, Scenario 1 (CSDC–CSDC) and Scenario 10 (DSSC–DSSC) are defined as baseline strategies, representing fully coordinated and fully decentralized configurations, respectively. These benchmarks allow for the performance of hybrid scenarios to be assessed against the extremes of centralized and decentralized decision-making structures. The subsequent tables (Tables 14–23) provide optimal values and solution parameters for selected scenarios, enabling analysis of how pricing, delivery times, and profit distributions vary across different competitive alignments.

The stopping condition was met after 6 repetitions.

In this scenario, both chains had identical strategic structures. However, the second chain did not offer a separate production line to serve customers of the higher class. Given that customers with the lower class were not very sensitive in this situation, the second chain generated more profit for customers in both classes despite having the same production capacity as the first chain.

The stopping condition was met after 8 repetitions

In this scenario, the strategic structures of the two chains were entirely distinct. One chain opted for a centralized strategy, while the other chose a decentralized or internal competition strategy. However, both chains adopted an exclusive production policy for each class of customers. In this situation, the first chain focused on its strategy and catering to customer demands more effectively, captured a larger market share, and generated more profit.

The stopping condition was met after 4 repetitions

In this scenario, it can be observed that chains operating with a centralized structure tend to generate more profit. Both chains performed poorly in this case, and the competition did not yield good profits, and both chains could have earned more substantial profits compared to the previous scenarios.

The stopping condition was met after 6 repetitions

As the chains had similar structures, their profits were minimal, and their differentiation was primarily derived from their internal costs.

The stopping condition was met after 6 repetitions

In this scenario, it was predicted that the first chain, which operated with a centralized structure and joint production, would generate more profit. This prediction was found to be consistent with the actual outcome.

The stopping condition was met after 4 repetitions

The stopping condition was met after 7 repetitions

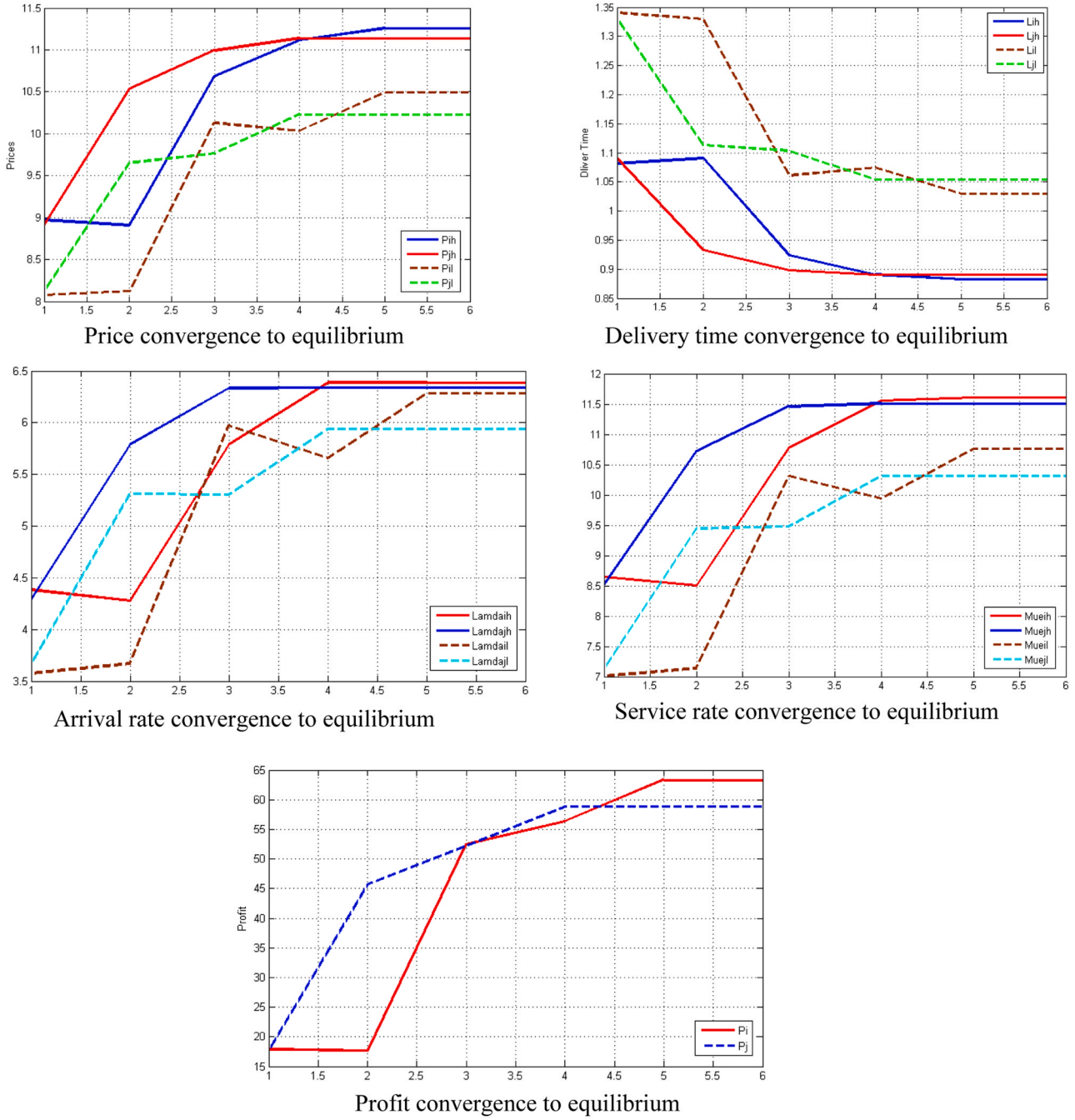


Fig. 13. Changes in parameters during the competition between CSDC-CSDC.

Table 14
Optimal solution values of the competition between CSDC-CSSC.

P_h^i	P_l^i	L_h^i	L_l^i	μ_h^i	μ_l^i	λ_h^i	λ_l^i	Π_{CSDC}^i
11.42	10.62	0.87	1.02	11.82	10.93	6.54	6.41	66.78
P_h^j	P_l^j	L_h^j	L_l^j	μ^j		λ_h^j	λ_l^j	Π_{CSSC}^i
11.47	10.07	0.47	0.99	16.14		6.41	6.40	72.33

As the chains had similar structures and minimal differentiation, they achieved comparable outcomes.

The stopping condition was met after 5 repetitions

In this given scenario, significant fluctuations and major changes occurred before the chains ultimately reached a state of equilibrium.

The stopping condition was met after 8 repetitions

After analyzing all competition scenarios, we consolidated the results, including the benefits of each chain, in Table 23 as follows:

Table 22 summarizes the profits of both supply chains under various competition scenarios. Each scenario represents a unique combination of competitive strategies, resulting in differing profit outcomes for both chains. Scenarios 1 (CSDC–CSDC) and 10 (DSSC–DSSC) are considered as benchmark strategies representing, respectively, full coordination and complete decentralization within and across supply chains. These serve as cooperative and non-cooperative baselines against which hybrid strategies can be assessed. Comparative analysis shows that hybrid configurations such as CSDC–CSSC or CSDC–DSSC offer up to 20 % improvement in total profit over the baselines. These improvements highlight the strategic advantage of mixed decision-making structures, especially under dual customer sensitivity settings.

Table 15
Optimal solution values in the competition between CSDC-DSDC.

P_h^i		P_l^i		L_h^i	L_l^i	μ_h^i	μ_l^i	λ_h^i	λ_l^i	Π_{CSDC}^i	
11.71		10.72		0.85	1.01	12.27	11.00	6.88	6.45	71.79	
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ_h^j	μ_l^j	λ_h^j	λ_l^j	$\Pi_{DSDC}^{Mnf_j}$	$\Pi_{DSDC}^{Dst_j}$
11.21	14.67	10.30	13.30	0.88	1.04	11.74	10.56	3.20	3.01	15.45	19.48

Table 16
Optimal solution values of the competition between CSDC-DSSC.

P_h^i		P_l^i		L_h^i	L_l^i	μ_h^i	μ_l^i	λ_h^i	λ_l^i	Π_{CSDC}^i	
9.53		8.56		1.02	1.26	9.42	7.75	4.91	4.08	26.02	
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ^j		λ_h^j	λ_l^j	$\Pi_{DSSC}^{Mnf_j}$	$\Pi_{DSSC}^{Dst_j}$
9.20	12.16	7.67	10.53	0.78	0.78	10.26		1.61	2.14	4.98	10.53

Table 17
Optimal solution values of the competition between CSSC-CSSC.

P_h^i	P_l^i	L_h^i	L_l^i	μ^i	λ_h^i	λ_l^i	Π_{CSSC}^i
9.29	7.74	0.63	1.49	11.60	4.26	4.08	25.68
P_h^j	P_l^j	L_h^j	L_l^j	μ^j	λ_h^j	λ_l^j	Π_{CSSC}^j
9.32	7.90	0.60	1.05	12.13	4.40	4.11	26.78

As the first chain acts as the market leader under the Stackelberg equilibrium, the following trends are observed regarding optimal cooperation-competition strategies and production capacity approaches for the second chain:

1. When the first chain employs the CSDC strategy, the second chain achieves the highest profit with the CSSC approach.
2. If the first chain chooses the CSSC strategy, the second chain benefits most from adopting the DSSC approach.
3. For a DSDC strategy by the first chain, the second chain maximizes its profit by also adopting the DSSC approach.
4. Similarly, when the first chain employs the DSSC strategy, the second chain's optimal choice remains the DSSC approach.

These findings indicate that the second chain consistently has opportunities to influence the market by selecting superior strategic combinations.

The analysis identifies the CSDC-DSSC combination as the most profitable scenario for both chains, yielding the highest total profit. This result underscores the significance of strategic decision-making and policy selection in optimizing profitability and achieving market dominance in competitive environments.

Table 18
Optimal solution values of the competition between CSSC-DSDC.

P_h^i		P_l^i		L_h^i	L_l^i	μ^i		λ_h^i	λ_l^i	Π_{CSSC}^i	
9.94		8.76		0.43	0.44	15.78		4.95	4.86	35.66	
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ_h^j	μ_l^j	λ_h^j	λ_l^j	$\Pi_{DSDC}^{Mnf_j}$	$\Pi_{DSDC}^{Dst_j}$
8.91	11.22	8.06	9.93	1.08	1.32	8.68	7.15	2.16	1.78	1.58	7.94

Table 19
Optimal solution values of the competition between CSSC-DSSC.

P_h^i		P_l^i		L_h^i	L_l^i	μ^i	λ_h^i	λ_l^i	Π_{CSSC}^i	
9.23		7.83		0.61	1.07	11.89	4.32	4.04	25.57	
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ^j	λ_h^j	λ_l^j	$\Pi_{DSSC}^{Mnf_j}$	$\Pi_{DSSC}^{Dst_j}$
9.03	9.03	7.50	7.50	0.79	0.79	10.01	4.17	5.05	9.22	33.80

Table 20
Optimal solution values of the competition between DSDC-DSDC.

P_{mh}^i	P_{dh}^i	P_{ml}^i	P_{dl}^i	L_h^i	L_l^i	μ_h^i	μ_l^i	λ_h^i	λ_l^i	$\Pi_{DSDC}^{Mnf_i}$	$\Pi_{DSDC}^{Dst_i}$
9.57	12.20	8.65	10.82	1.01	1.23	9.59	8.05	2.46	2.10	2.01	10.59
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ_h^j	μ_l^j	λ_h^j	λ_l^j	$\Pi_{DSDC}^{Mnf_j}$	$\Pi_{DSDC}^{Dst_j}$
9.57	12.20	8.65	10.82	1.01	1.23	9.59	8.05	2.46	2.10	2.01	10.59

Table 21
Optimal solution values of the competition between DSDC-DSSC.

P_{mh}^i	P_{dh}^i	P_{ml}^i	P_{dl}^i	L_h^i	L_l^i	μ_h^i	μ_l^i	λ_h^i	λ_l^i	$\Pi_{DSDC}^{Mnf_i}$	$\Pi_{DSDC}^{Dst_i}$
9.27	11.76	7.81	9.57	1.03	1.38	9.43	6.50	2.44	1.53	8.19	8.36
P_{mh}^j	P_{dh}^j	P_{ml}^j	P_{dl}^j	L_h^j	L_l^j	μ^j		λ_h^j	λ_l^j	$\Pi_{DSSC}^{Mnf_j}$	$\Pi_{DSSC}^{Dst_j}$
9.64	9.64	7.95	7.95	0.75	0.75	10.96		4.79	5.48	43.61	10.28

Table 22
Optimal solution values of the competition between DSSC-DSSC.

P^i_{mh}	P^i_{ml}	P^i_{dh}	P^i_{dl}	L^i_h	L^i_l	μ^i	λ^i_h	λ^i_l	$\Pi^{Mnf_i}_{DSSC}$	$\Pi^{Dst_i}_{DSSC}$
9.08	7.44	12.37	7.44	0.79	0.79	10.11	7.05	6.25	13.22	6.25
P^j_{mh}	P^j_{ml}	P^j_{dh}	P^j_{dl}	L^j_h	L^j_l	μ^j	λ^j_h	λ^j_l	$\Pi^{Mnf_j}_{DSSC}$	$\Pi^{Dst_j}_{DSSC}$
9.57	7.57	9.57	7.57	0.76	0.76	10.89	4.86	4.93	39.07	9.79

Table 23
Chains profits in all competition scenarios.

		Profit of chain i			Profit of chain j		
Scenario	Competition	Mnf.	Dst.	Whole	Mnf.	Dst.	Whole
1	CSDC - CSDC			63.33			58.79
2	CSDC - CSSC			66.78			72.33
3	CSDC - DSDC			71.79	15.45	19.48	34.93
4	CSDC - DSSC			26.02	4.98	10.53	15.51
5	CSSC - CSSC			25.68			26.78
6	CSSC - DSDC			35.66	1.58	7.94	9.52
7	CSSC - DSSC			25.57	9.22	33.8	43.02
8	DSDC - DSDC	2.01	10.59	12.6	2.01	10.59	12.6
9	DSDC - DSSC	8.19	8.36	16.55	43.61	10.28	53.89
10	DSSC - DSSC			63.33			58.79

5.4. Robustness and model validation

To ensure the reliability and credibility of the proposed game-theoretic model, several robustness checks and validation measures were conducted. The convergence behavior of the Stackelberg-based solution algorithm, detailed in Section 5.2, was examined under varying initial conditions. Across all simulations involving the three strategic configurations (CSDC, DSSC, and Hybrid), the model consistently achieved stable equilibrium outcomes, regardless of the initial values for price and delivery time. This outcome demonstrates that the iterative optimization process reliably converges to Stackelberg equilibrium points. Furthermore, to confirm the independence of the key behavioral coefficients introduced in Section 3 namely, the responsiveness coefficient (γ), environmental volatility (β), and internal coordination sensitivity (θ) the Variance Inflation Factor (VIF) test was applied. All parameters exhibited VIF values below 2.5, indicating minimal multicollinearity and confirming the distinct and interpretable influence of each parameter on customer behavior.

To evaluate the model's stability under stochastic variation, multiple replications of the numerical experiments were performed using randomly initialized parameter values within realistic ranges identified in the literature. These variations included adjustments to customer arrival rates, service capacity, and initial pricing assumptions. The resulting equilibrium prices, delivery times, market shares, and profit values remained consistent across runs, demonstrating the model's insensitivity to input fluctuations and its internal coherence. Additionally, the ten strategic scenarios outlined in Section 4.1 were validated based on realistic coordination structures and supported by prior studies [8,32]. This validation confirms that the scenario design effectively captures diverse competitive dynamics while maintaining practical relevance. Collectively, these steps reinforce the model's robustness, solution reliability, and practical applicability to decision-making in dual supply chain environments.

6. Conclusion

6.1. Summary of findings

This study develops a game-theoretic model to analyze competition between two supply chains serving distinct customer segments. The numerical results demonstrate that the CSDC–DSSC strategy combination yields the highest total profit for both chains, underscoring the importance of strategic alignment in competitive environments. Notably, the second chain consistently achieves better profitability when it complements the first chain's strategy for instance, by choosing CSSC when the first chain selects CSDC, or DSSC in response to other strategies. These outcomes highlight the critical role of dynamic and adaptive decision-making in maximizing overall market performance. Additionally, the findings reveal that while dedicated production lines for high-class customers increase satisfaction and short-term gains, shared production strategies may offer superior cost-efficiency for price-sensitive segments, particularly when separation costs are high. These insights provide a foundation for understanding the trade-offs between service differentiation and operational efficiency.

6.2. Managerial implications

The results offer actionable insights for supply chain managers across various industries. Companies operating in time-sensitive sectors such as express logistics, pharmaceuticals, and premium e-commerce are advised to adopt centralized-dedicated strategies (e.g., CSDC–CSDC), which facilitate coordinated pricing and reliable delivery performance for high-value customers. In contrast, hybrid strategy combinations such as CSDC–CSSC or CSSC–DSSC strike a balance between responsiveness and cost efficiency, particularly under conditions of demand uncertainty or mixed customer priorities. The model also suggests that partial decentralization, when paired with customer segmentation, can enhance profitability without entirely compromising control or service quality. These findings empower managers to configure supply chain strategies that align with both operational capabilities and market expectations.

6.3. Future research

Future research can extend the current model by incorporating additional customer segments, multi-echelon supply chain tiers, or stochastic demand scenarios. Exploring dynamic pricing, real-time coordination mechanisms, and environmental constraints such as carbon emissions could further enrich the analysis. These extensions would offer broader applicability and help organizations fine-tune their strategic responses to complex and evolving market environments.

CRediT authorship contribution statement

Masoud Narenji: Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Armin Mahmoodi:** Writing –

original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Seyed Mojtaba Sajadi:** Supervision, Investigation, Funding acquisition, Conceptualization. **Maryam Amini:** Funding acquisition, Formal analysis, Data curation, Conceptualization. **Milad Jasemi:** Writing – original draft, Validation, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1

Before delving into the modeling and problem-solving aspects, exclusive and joint production capacity was discussed. In this scenario, the producer has two classes of customers and the option to choose a shared capacity policy, which would involve using one production line for each class of customers and following an $M/M/1$ queue model. However, some producers prefer to provide better service to customers of a higher class. In such cases, a dedicated capacity policy may be more effective in offering faster service, excellent reliability, and at a different cost. In the following sections, an overview of the waiting time limit for customers in the queue is provided, and its distribution function is extracted for two modes of production - joint and exclusive.

Delivery time reliability for both modes is derived analytically under stable conditions. In dedicated production mode, the distribution of waiting time in the queue is as follows:

$$S_k^i(L_k^i) = P(W_k^i \le L_k^i) = 1 - \quad e^{(\lambda_k^i - \mu_k^i)L_k^i} \ge \alpha$$

In the case of shared capacity production, customers with any priority arrive at a single-channel server with input rate λ_k^i, and wait first-in, first out for service in their respective priority area. Calculating the waiting time in the queue for this mode is a complex problem. Previous studies such as Stephan, Cohen, and Heyman and Sobel (1982) have used Laplace transforms to calculate the distribution of waiting time and provided numerical examples. However, some approximate methods other authors propose, such as Abate and Whitt (1997), need to be more accurate. One of the most appropriate methods for calculating the waiting time in the queue for two classes of customers is the method of geometric planning and cutting pages, which will be explained later.

To simplify the problem of waiting time distribution in collaborative production mode without losing generality, we will focus on only one chain, $S_l^k(.)$, representing the lower class of customers. The calculation details are like the methods provided by Neuts and Nelson.

We can represent the number of high-class and low-class customers at time t as $N_h(t)$ and $N_l(t)$, respectively. Therefore, $\{N(t)\} := \{N_l(t), N_h(t), t > 0\}$ is a two-dimensional continuous-time Markov chain with state space $\{n = (n_l, n_h)$. By solving the geometric matrix, we can use $N(t)$as a production-death process to obtain the distribution of customers in each class. To achieve this, we assume that the length of the high-priority customer queue (including the customer receiving service) is limited to a maximum of M. This value should be large enough to ensure accurate results.

In the Markov process, $\{N(t)\}$, a transition can occur when a customer from either class enters, or a customer from either class receives service. All possible states for the Markov process are shown in Table 23, where λ_h and λ_l are known input rates for high and low-priority customers, respectively.

Table 23
Rate of Change for Priority Queues

From	To	rate of change	Conditions
(n_l, n_h)	$(n_l, n_h + 1)$	λ_h	$n_l \ge 0$, $n_h \ge 0$
(n_l, n_h)	$(n_l + 1, n_h)$	λ_l	$n_l \ge 0$, $n_h \ge 0$
(n_l, n_h)	$(n_l, n_h - 1)$	μ	$n_l \ge 0$, $n_h > 0$
(n_l, n_h)	$(n_l - 1, n_h)$	μ	$n_l > 0$, $n_h = 0$

$$Q = \begin{pmatrix} B_0 & A_0 & & & \\ A_2 & A_1 & A_0 & & \\ & A_2 & A_1 & A_0 & \\ & & \ddots & \ddots & \ddots \\ & & & & \end{pmatrix}$$

The table shows the rate of change for priority queues. Here, B_0‘ A_0‘ A_1 and A_2 are square matrices with dimensions of $M + 1$. We can rewrite these matrices as follows:

$$A_0 = \begin{pmatrix} \lambda_l & & & & \\ & \lambda_l & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \lambda_l \end{pmatrix}$$

$$A_2 = \begin{pmatrix} \mu & & & & \\ & 0 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 0 \end{pmatrix}$$

$$B_0 = \begin{pmatrix} * & \lambda_h & & & \\ \mu & * & \lambda_h & & \\ & \mu & * & \lambda_h & \\ & & \ddots & \ddots & \ddots \\ & & & \mu & * \end{pmatrix}$$

The value of $*$ is obtained by solving the equation $A_0 e + B_0 e = 0$. We can also obtain the value of A_1 from the equation $A_1 = B_0 - A_2$. The value of X as a probability vector for $\{N(t)\}$ is calculated as follows:

$$X = [x_{00},\ x_{01}, \ldots,\ x_{0M},\ x_{10},\ x_{11},\ \ldots,\ x_{1M},\ \ldots,\ \ldots,\ x_{i0},\ x_{i1},\ \ldots,\ x_{iM},\ \ldots,\ \ldots\ ]$$

We can partition vector X into smaller vectors X_i where $X_i = [x_{i0},\ x_{i1},\ \ldots,\ x_{iM}]$. Thus, X can be expressed as $X = [X_0,\ X_1,\ X_3,\ \ldots,\ \ldots\ ]$. The equilibrium equations for X can be represented in the standard matrix form, which can be written as a set of equations. This form has been extensively discussed in the literature, including works by Latouche and Ramaswami (1999), Nelson, and Neuts:

$$XQ = 0; X_{i+1} = X_i R$$

Where R is obtained by solving the following second-degree equation:

$$A_0 + RA_1 + R^2 A_2 = 0$$

The matrix R can be obtained using a well-known method (Latouche and Ramaswami, 1999). This method involves an iterative procedure for calculating R:

$$R(0) = 0; R(n+1) = -\left[A_0 + R^2(n)A_2\right]A_1^{-1}$$

We can obtain the value of X_0 from the following equation:

$$X_0 = (B_0 + RA_2) = 0$$

The normalized form of the above equation is as follows:

$$\sum_{i=0}^{\infty} X_i e = X_0 (I - R)^{-1} e = 1$$

Here, a is a vector with a dimension of $M + 1$.

The waiting time W_l for a low-priority customer is the time between arrival and service completion. This waiting time can be affected by the number of customers of both classes who have arrived before them and by the number of high-priority customers who arrive after them and receive service before them. To calculate the waiting time of a specific low-priority customer, we need to consider the waiting time for that customer and the waiting time caused by a modulated Markov chain $\{\widetilde{N}(t)\}$ with an absorption rate of $\lambda_l = 0$. We assume that the matrix $\widetilde{A}_0$, which shows the entry and exit of high-priority customers, is zero. The state of absorption is represented by $0'$, which is the state in which a specific customer has finished receiving their service. The following matrix represents the unbounded estimate for this process:

$$\widetilde{Q} = \left(\begin{array}{c|ccc} 0 & 0 & 0 & \ldots \\ b_0 & \widetilde{B}_0 & 0 & \ldots \\ 0 & A_2 & \widetilde{A}_1 & \\ 0 & & A_2 & \widetilde{A}_1 \\ \vdots & & \ddots & \ddots \end{array}\right)$$

where $\widetilde{B}_0 = B_0 + A_0$‘ $\widetilde{A}_0 = A_1 + A_0$ and $b_0 = [\mu 00 \ldots 0]^T_{M+1} + 1$. The first row and column in the above matrix represent the absorption of the $0'$ state. The time spent for a particular customer, which follows the adjusted Markov process with the rate Q˜, has a direct relationship with the price of the products ($P_h,\ P_l$). Prices also determine the entry rate ($\lambda_h,\ \lambda_l$(and service μ. For prices (P and Royce rate μˆk, the waiting time distribution function of low-priority customers is represented by $S_l^k(y) = 1 - \overline{S_l^k}(y)$, which is It $\overline{S_l^k}(y)$ indicates that the probability of the waiting time of a customer with a low priority in the system is more than y units. We also assume that $\overline{S_{li}^k}(y)$ is the probability that a particular customer who sees low priority customer i in front of him spends more than y units of time in the system. In his article, Mr. (Wolff 1982) calculated the probability that a particular customer will see low priority customer i in front of him. Having the relationship $X_i = X_0 R^i$, the value of $\overline{S_l^k}(y)$ can be represented as follows:

$$\overline{S_l^k}(y) = \sum_{i=0}^{\infty} X_i \overline{S_{li}^k}(y) e$$

The value of $\overline{S_{li}^{k}}(y)$can be rewritten using the Markov process $\{\widetilde{N}(t)\}$ and Poisson process with rate γ. that the arrival rate of Powasan is equal to:

$$\gamma = \max_{0\le i\le M}(-\widetilde{A}_1)_{ii} = \max_{0\le i\le M} -(A_0+A_1)_{ii}$$

Therefore, the matrix $\widetilde{Q}$ can be rewritten as a probability matrix with discrete time as follows:

$$\widehat{Q} = \frac{1}{\gamma}\widetilde{Q} + I = \left(\begin{array}{c|ccc} 1 & 0 & 0 & \cdots \\ \widehat{b}_0 & \widehat{B}_0 & 0 & \cdots \\ 0 & \widehat{A}_2 & \widehat{A}_1 & \\ 0 & & \widehat{A}_2 & \widehat{A}_1 \\ \vdots & & \ddots & \ddots \end{array}\right)$$

where $\widehat{A}_2 = \frac{A_2}{\gamma}$، $\widehat{A}_1 = \frac{\overline{A}_2}{\gamma} + I$ and $\widehat{b}_0 = \frac{b_0}{\gamma}$. The probability that n number of Poisson events will occur at time y equals $e^{-\gamma y}\frac{(\gamma y)^n}{n!}$. Let us assume that a particular customer sees i customer with low priority in front of him, in this case, for his waiting time in the queue to exceed y, a maximum of i customers out of n customers with low priority should be moved with customers with high priority. In other words:

$$\overline{S_{li}^{k}}(y) = \sum_{n=0}^{\infty} e^{-\gamma y}\frac{(\gamma y)^n}{n!}\sum_{v=0}^{i} G_v^{(n)} e, \quad i \ge 0$$

Where $G_v^{(n)}$ is the matrix?

whose elements are the conditional probabilities that make n changes with a discrete-time Markov process and rate $\widehat{Q}$. Also, there are v cases of changes related to customers with low priority. By substituting the above expression, we will have:

$$\overline{S_{l}^{k}}(y) = \sum_{n=0}^{\infty} d_n e^{-\gamma y}\frac{(\gamma y)^n}{n!}$$

Where d_n is equal to:

$$d_n = \sum_{i=0}^{\infty} X_0 R^i \sum_{v=0}^{i} G_v^{(n)} e, \quad n \ge 0$$

By simplifying the Foy expression, we will have:

$$\sum_{i=0}^{\infty} R^i \sum_{v=0}^{i} G_v^{(n)} e$$

$$= \sum_{i=0}^{n+1} R^i \sum_{v=0}^{i} G_v^{(n)} e + \sum_{i=n+2}^{\infty} R^i \sum_{v=0}^{n} G_v^{(n)} e$$

while for υ > n the value of $G_v^{(n)} = 0$.

$$= \sum_{v=0}^{n+1}\sum_{i=v}^{n+1} R^i G_v^{(n)} e + (I-R)^{-1} R^{n+1} e$$

Since the $\sum_{v=0}^{n} G_v^{(n)} e = e$

$$= \sum_{v=0}^{n+1} (I-R)^{-1}\left(R^v - R^{n+2}\right) G_v^{(n)} e + (I-R)^{-1} R^{n+2} e$$

$$= \sum_{v=0}^{n} (1-R)^{-1} R^v G_v^{(n)} e + (I-R)^{-1} R^{n+1} G_{n+1}^{(n)} e$$

while $\sum_{v=0}^{n+1} G_v^{(n)} e = e$

$$= \sum_{v=0}^{n} (I-R)^{-1} R^v G_v^{(n)} e$$

Considering that for υ > n the value of $G_v^{(n)} = 0$.

$= (I-R)^{-1} H_n e$ $n \ge 0$that the value of H_n is equal to $\sum_{v=0}^{n} R^v G_v^{(n)}$.

Therefore:

$$S_l^k(L_l) = 1 - \overline{S_l^k}(L_l) = \sum_{n=0}^{\infty} e^{-\gamma L_l}\frac{(\gamma L_l)^n}{n!} X_0 (I-R)^{-1} H_n e$$

$$S_l^k(L_l) = 1 - \overline{S_l^k}(L_l) = \sum_{n=0}^{\infty} e^{-\gamma L_l} \frac{(\gamma L_l)^n}{n!} X_0 (I - R)^{-1} H_n e \tag{39}$$

The value of H_n can also be calculated as the following regression equation:

$$H_{n+1} = H_n \widehat{A}_1 + R H_n \widehat{A}_2; H_0 = I$$

Therefore, having the values (P_h^k, P_l^k(and the service rate) μ^k), the value of $S_l^k(.)$ can be calculated using the Eq. (39).

Appendix 2

Proof of Theorem 1: To show that the model is strictly concave, we must demonstrate that its Hessian matrix is negative definite. The Hessian matrix of the objective function is as follows:

$$H = \begin{bmatrix} -2\left(\beta_P^h + \theta_p + \gamma_p\right) & 2\theta_p & -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L \\ 2\theta_p & -2\left(\beta_P^l + \theta_p + \gamma_p\right) & \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) \\ -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L & 0 & 0 \\ \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) & 0 & 0 \end{bmatrix}$$

Upon performing the necessary calculations, we obtain the determinant of the Hessian matrix as

$$-\theta_L \left\{ \left(\beta_L^h + \theta_p + \gamma_p\right)\left(\beta_L^l + \theta_p + \gamma_p\right)\left(\beta_L^l + 2\theta_p + \gamma_p\right) + 2(\theta_L)^2 \right\}$$

Proof of Theorem 2: To prove the above theorem, we used data from Table 2 and graphically demonstrate the convexity of the distribution function. As shown in the Figs. 14–15, the waiting time distribution is indeed convex for the price and service rate parameters:

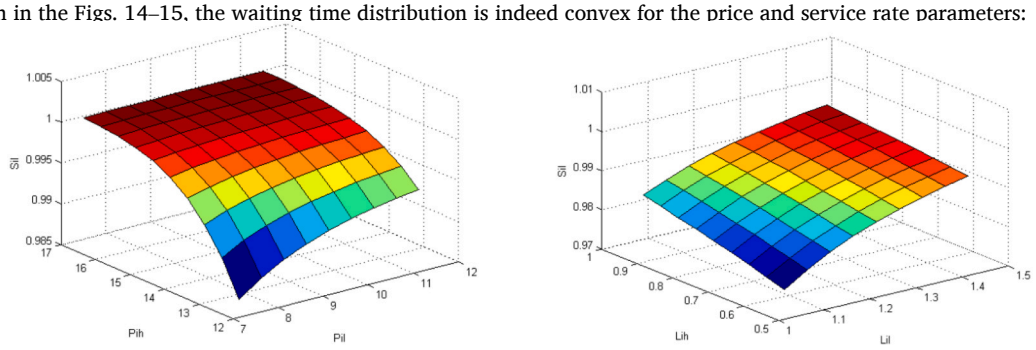


Fig. 14. Effect of prices and delivery times of chain reliability for low-class customers

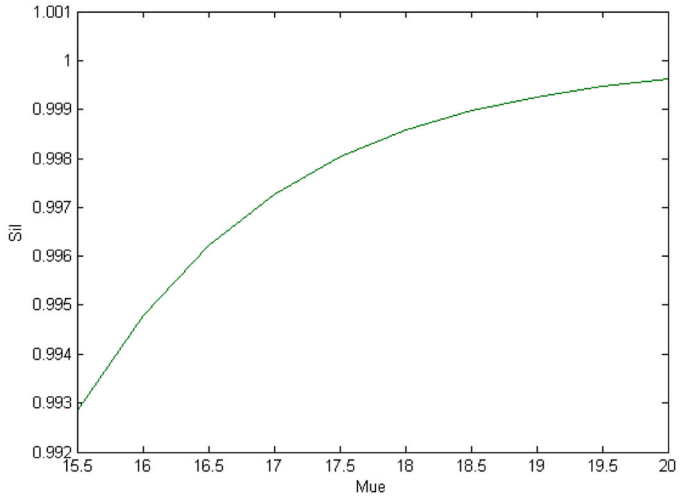


Fig. 15. Effect of service rate on chain reliability for low-class customers

As the service rate increases, the supply chain reliability also increases, and the probability of distribution of waiting time for low-class customers becomes closer to 1. High-class customers, having priority and receiving service as soon as they enter the system, have a higher level of reliability, and their probability of distribution of waiting time is significantly better.

In the CSSC model, the objective function is a nonlinear function, and the constraints, except for Eq. (13), are either linear or nonlinear but known. However, the distribution of Eq. (13), which is related to supply chain reliability, still needs to be discovered. Since the probability distribution of waiting time for low-class customers is convex concerning the price parameters of both classes, delivery times for both classes, and the service rate, it is

possible to estimate this relationship with $S_l^i(.) = \min_{k\epsilon K} S_l^{ik}(.) + \left(P_h^i - P_h^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial P_h^i}\right) + \left(P_l^i - P_l^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial P_l^i}\right) + \left(L_h^i - L_h^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial L_h^i}\right) + \left(L_l^i - L_l^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\partial L_l^i}\right) + \left(\mu^i - \mu^{ik}\right)\left(\frac{\partial S_l^{ik}(.)}{\mu^i}\right)$.

Proof of Theorem 3: We form the Hessian matrix for the objective function of the problem.

$$H = \begin{array}{|c|c|c|c|c|} \hline -2\left(\beta_P^h + \theta_p + \gamma_p\right) & 2\theta_p & -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L & 0 \\ \hline 2\theta_p & -2\left(\beta_P^l + \theta_p + \gamma_p\right) & \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) & 0 \\ \hline -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L & 0 & 0 & 0 \\ \hline \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

Calculating the matrix's determinant above can easily prove the results' uniqueness.

Proof of Theorem 4. involves constructing the Hessian matrix for the objective function of the problem.

$$H_{DSDC}^{Dst-i} = \begin{bmatrix} -2\left(\beta_P^h + \theta_p + \gamma_p\right) & 2\theta_p \\ 2\theta_p & -2\left(\beta_P^l + \theta_p + \gamma_p\right) \end{bmatrix}$$

$$H_{DC-D}^{Mnf-i} = \begin{bmatrix} 0 & 0 & -\left(\beta_L^h + \theta_L + \gamma_L\right) & \theta_L & 0 & 0 \\ 0 & 0 & \theta_L & -\left(\beta_L^h + \theta_L + \gamma_L\right) & 0 & 0 \\ -\left(\beta_L^h + \theta_L + \gamma_L\right) & \theta_L & 0 & 0 & 0 & 0 \\ \theta_L & -\left(\beta_L^h + \theta_L + \gamma_L\right) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The above matrices, obtained by calculating the determinants, demonstrate a unique solution to the problem.

Proof of Theorem 5: The Hessian matrix is formed for the objective function of the problem.

$$H = \begin{array}{|c|c|c|c|c|} \hline -2\left(\beta_P^h + \theta_p + \gamma_p\right) & 2\theta_p & -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L & 0 \\ \hline 2\theta_p & -2\left(\beta_P^l + \theta_p + \gamma_p\right) & \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) & 0 \\ \hline -\left(\beta_L^h + \theta_p + \gamma_p\right) & \theta_L & 0 & 0 & 0 \\ \hline \theta_L & -\left(\beta_L^l + \theta_p + \gamma_p\right) & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

The uniqueness of the results has been easily proven by calculating the determinant of the above matrix.

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